



Seminar: Monte Carlo Methods in Finance and Econometrics (F)

Monte Carlo Pricing of Asian Options Under The CEV Model:
Variance Reduction and Sensitivity Analysis

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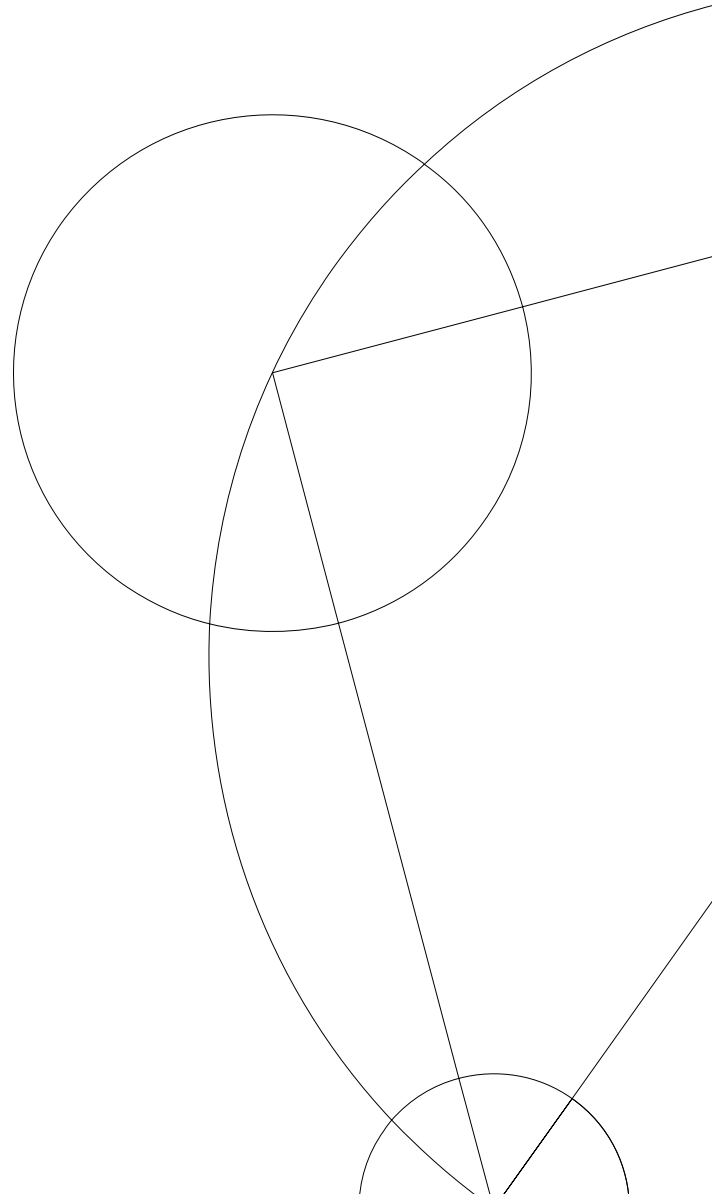
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Abstract

This seminar project examines the pricing of arithmetic Asian call options under the Constant Elasticity of Variance (CEV) model. Because the payoff depends on the entire price path and there is no closed-form solution under CEV dynamics, option prices must be approximated numerically. We compare standard Crude Monte Carlo (CMC) with Randomized Quasi-Monte Carlo (RQMC) methods based on scrambled Sobol' sequences, and study how Brownian bridge construction and variance reduction techniques affect accuracy and efficiency. The CEV process is simulated using Euler–Maruyama and Milstein schemes, with Milstein used in the main analysis. We find that RQMC combined with Brownian bridge construction delivers substantial variance reductions and efficiency gains relative to CMC. Control variates improve CMC but are less effective for RQMC, while antithetic variates perform poorly when combined with RQMC. The finite-difference estimates of Delta and Vega show that Greek estimation is more noise-sensitive than pricing, while RQMC still delivers substantial variance reductions. Overall, the results show that advanced Monte Carlo methods can greatly improve the pricing of path-dependent options under the CEV model, while also revealing that improvements in pricing efficiency do not always translate into more accurate sensitivity estimates.

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1 Introduction

Monte Carlo methods have become an essential tool in modern financial engineering, particularly for pricing complex derivatives where analytical solutions are unavailable. One of such derivatives is the arithmetic Asian option, which is often traded on markets for crude oil, metals, interest rates, and currencies, for hedging against short-term volatility and market manipulation. Unlike standard European options, their payoffs depend on the average of the underlying asset prices over time, which is why they are typically referred to as path dependent. However, path dependence makes their valuation analytically impossible to calculate under most stochastic models, which is why we need Monte Carlo simulations to estimate an appropriate price for such an option.

In our paper, we will employ the Constant Elasticity of Variance (CEV) model, first introduced by J. Cox (1975), which extends the classical Black–Scholes framework by allowing the volatility of the underlying asset to vary with its price level. This feature captures empirical phenomena such as the leverage effect and implied volatility skew observed in real markets, making the CEV model an appealing alternative for option pricing. However, the absence of a closed-form pricing formula for arithmetic Asian options under the CEV dynamics necessitates numerical approximation techniques such as Monte Carlo simulations.

Despite its flexibility, the crude Monte Carlo (CMC) estimator converges slowly at a rate of $O(n^{-1/2})$ making it computationally expensive to achieve high precision. This has motivated the development of variance reduction and quasi–Monte Carlo (QMC) methods, which aim to improve efficiency by reducing estimator variance. QMC methods replace pseudo-random draws with low-discrepancy sequences that provide a more uniform coverage of the integration space, while randomization techniques, such as scrambling, allow for error estimation and statistical inference. Further improvements can be achieved by incorporating Brownian bridge construction, which reduces the effective dimensionality of the problem.

In this project, we investigate how different Monte Carlo variants perform in pricing arithmetic Asian options under the CEV model. Specifically, we compare the performance of CMC and randomized QMC (RQMC) methods, assess the impact of Brownian bridge construction, and implement classical variance reduction techniques such as control variates and antithetic variates. Furthermore, we extend the analysis by estimating the sensitivities (Greeks) of the option value with respect to key model parameters, focusing on Delta and Vega.

Our goal is to evaluate and contrast the accuracy, convergence behaviour, and computational efficiency of these simulation methods. Through numerical experiments, we demonstrate how advanced simulation techniques can significantly improve the precision

of Monte Carlo estimators for path-dependent options. More formally, we show that randomized quasi-Monte Carlo methods combined with Brownian bridge construction and a control variate, lead to substantial efficiency gains compared to standard Monte Carlo for pricing path-dependent options under the CEV model. However, for Greek estimation, the use of variance reduction techniques on RQMC impacts the estimation negatively.

The remainder of the paper is organized as follows. Section 2 introduces the Asian option pricing problem. Section 3 presents the CEV model dynamics and discretization schemes. Sections 4–6 describe quasi-Monte Carlo methods, variance reduction techniques and the estimation of Greeks. Section 7 presents the numerical results, while Section 8 contains the sensitivity analysis. Section 9 concludes.

2 The Asian Option Pricing Problem

An Asian option is an exotic option that gives the holder, the right to buy or sell an asset for a given strike price K , where the payoff is determined by the average price of the asset in a specified period of time. This makes Asian options path-dependent, as the payoff is determined by the trajectory of the underlying asset, as opposed to an European option, where the payoff is determined only by the price of the asset at the end of the options expiration and the strike price. This averaging feature makes Asian options less sensitive to short-term price fluctuations and thereby reduces their exposure to market manipulation or extreme volatility close to maturity.

However, as the average can be calculated multiple ways, the common Asian options are usually categorized as either an arithmetic or geometric Asian option. The Arithmetic Asian call option's payoff, with discrete monitoring is calculated as $(\bar{S} - K)^+$, where

$$\bar{S} = \frac{1}{M} \sum_{i=1}^M S(t_i) \quad (1)$$

Where $S(t_i)$ denotes the price of the asset at time t_i . The geometric $\bar{S}$ is calculated as

$$\bar{S}_{geo} = \left(\prod_{i=1}^M S(t_i) \right)^{\frac{1}{M}} \quad (2)$$

Under Black-Scholes, the geometric Asian option does have a closed-form pricing solution, however the arithmetic does not and simulation based solutions is required. In this paper we consider the CEV model for our underlying asset, and so a closed-form solution for both the arithmetic and geometric options is non-existent, and simulation-based methods, such as Monte Carlo simulations is needed (Glasserman 2003, p.99).

For an arithmetic Asian call option with strike price K , the payoff at maturity is $(\bar{S} - K)^+$. Working under the risk-neutral measure Q , the arbitrage-free price of the option is therefore the discounted expectation of this payoff

$$P = e^{-rT} \mathbb{E}_Q[(\bar{S} - K)^+] \quad (3)$$

This formulation makes clear that pricing the option reduces to computing a single expectation under the risk-neutral distribution of the price path. Because the distribution of $\bar{S}$ under the CEV dynamics has no tractable form, the expectation must be evaluated numerically. Consequently, in the remainder of the paper we focus on Monte Carlo and quasi-Monte Carlo methods for approximating this expectation efficiently and accurately.

3 CEV model dynamics

The Constant Elasticity of Variance (CEV) model is particularly appealing in settings where volatility is known to depend on the asset level. This is especially relevant for Asian options, which are frequently written on commodities. Commodity prices commonly exhibit the inverse leverage effect, where volatility increases as the price rises. This behaviour is inconsistent with the constant-volatility assumption of the Black-Scholes model. The CEV specification captures this effect while keeping the model easy to work with and straightforward to simulate, making it well suited for the pricing problem considered in this project.

The CEV model is given by the stochastic differential equation (SDE)

$$dS_t = \mu S_t dt + \sigma S_t^\gamma dW_t \quad (4)$$

Where μ is the drift parameter, $\sigma > 0$ is the diffusion/volatility coefficient, γ is the elasticity parameter and W_t denotes a standard Brownian motion. For pricing purposes we work under a risk-neutral measure Q . By the fundamental theorem of asset pricing, the discounted price $f(t, S_t) = e^{-rt} S_t$ must be a Q -martingale, which implies that its drift under Q must equal the risk-free rate r . Applying Itô's lemma to $f(t, S_t)$ yields the risk-neutral dynamics

$$dS_t = r S_t dt + \sigma S_t^\gamma dW_t^Q. \quad (5)$$

Formally, this corresponds to a change of measure from the physical measure P to Q , which can be obtained via Girsanov's theorem. Under Q we therefore set $\mu = r$. γ determines how the volatility scales with the asset price. When $\gamma = 1$, the model reduces to the Black-Scholes model with constant volatility. For $\gamma > 1$ volatility increases as the asset price increases, which is consistent with the inverse leverage effect typically observed in commodity markets (Glasserman 2003, p. 133).

We assume that the CEV process admits an equivalent martingale measure under the tested parameter configuration. In particular, for the chosen elasticity parameter γ and volatility level σ , the process is assumed to stay strictly positive almost surely. While it is well known that the CEV process may hit zero for certain parameter regions, in this study this issue is mitigated numerically through the use of logarithmic transformations and sufficiently fine time discretization. Boundary behaviour at zero is therefore not examined explicitly in this work.

The instantaneous volatility of the returns is $\sigma S_t^{\gamma-1}$. This structure allows the CEV model to generate implied volatility skews that align more closely with market data than the flat volatility surface implied by Black–Scholes.

3.1 Discretization

Since the CEV model is continuous, simulation methods must rely on numerical discretization of the SDE. In this project, we consider both the Euler–Maruyama and the Milstein schemes for simulating the process. If we apply Ito’s Lemma to the logarithm of the asset price $X_t = \log S_t$, we get the SDE on a form that is more stable for simulation and guarantees positive asset values.

$$dX_t = \left(r - \frac{1}{2} \sigma^2 S_t^{2(\gamma-1)} \right) dt + \sigma S_t^{\gamma-1} dW_t = \left(r - \frac{1}{2} \sigma^2 e^{2(\gamma-1)X_t} \right) dt + \sigma e^{(\gamma-1)X_t} dW_t \quad (6)$$

The Euler–Maruyama scheme is simple and widely used for approximating stochastic differential equations. The continuous-time dynamics of the process are discretized on $[0, T]$ using an equidistant time grid $0 = t_0 < t_1 < \dots < t_M = T$ with constant step size $\Delta t = \frac{T}{M}$ over which the drift and diffusion terms are assumed to remain constant. Applied to the logarithmic CEV process dX_t the Euler–Maruyama update is given by

$$\tilde{X}_{i+1}^{euler} = \tilde{X}_i + \left(r - \frac{1}{2} \sigma^2 e^{2(\gamma-1)\tilde{X}_i} \right) \Delta t + \sigma e^{(\gamma-1)\tilde{X}_i} \Delta W_{i+1} \quad (7)$$

Where under standard MC, $\Delta W_{i+1} = \sqrt{\Delta t} Z_{i+1}$ with $Z_{i+1} \sim N(0, 1)$ being independent standard normal random variables (Glasserman 2003, p. 339). The simulated asset price can be recovered as $\tilde{S}_{i+1} = e^{\tilde{X}_{i+1}}$, which ensures positivity of all simulated paths. This method has a strong order of convergence $\mathcal{O}(\sqrt{\Delta t})$ and weak order $\mathcal{O}(\Delta t)$. Although it is simple and computationally efficient, the Euler–Maruyama scheme can accumulate discretization errors, especially when the diffusion term depends non-linearly on the state variable, as is in the case in the CEV model. Consequently, for small values of S_t or large σ , the approximation may become less stable, which motivates higher order schemes like the Milstein method.

The Milstein method refines the Euler–Maruyama approximation by including an additional term derived from Itô’s lemma that accounts for the derivative of the diffusion coefficient. This correction improves the accuracy of the discretization for SDEs with state-dependent volatility. The Milstein’s update for the logarithmic CEV process is given by

$$\tilde{X}_{i+1}^{milstein} = \tilde{X}_{i+1}^{euler} + \frac{1}{2}\sigma^2(\gamma - 1)e^{2(\gamma-1)\tilde{X}_i}((\Delta W_{i+1})^2 - \Delta t) \quad (8)$$

Where $\tilde{X}_{i+1}^{euler}$ is the term found in equation 7. The additional terms captures part of the curvature of the diffusion function, reducing bias in the simulation. The Milstein scheme achieves a strong order of convergence $\mathcal{O}(\Delta t)$, making it significantly more accurate than Euler–Maruyama for the same step size. However, the computational cost per step is slightly higher, as it involves an additional evaluation of the diffusion derivative (Glasserman 2003, p. 343).

The algorithm for simulating a single path of the CEV price process using either Euler–Maruyama or Milstein can be found in Appendix A1. In the subsequent analysis, we employ both discretization schemes to simulate the CEV process and assess their effect on the accuracy and variance of Monte Carlo estimates.

4 Quasi-Monte Carlo

An alternative to standard Monte Carlo (MC) simulation is quasi-Monte Carlo (QMC) simulation. QMC employs low-discrepancy sequences instead of pseudo-random draws to produce more uniformly spread sample points. The motivation is that, for many integrands, this can accelerate convergence from the MC rate $O(n^{-1/2})$ toward rates close to $O(\frac{1}{n^{1-\epsilon}})$ for any $\epsilon > 0$. The following section is based on Glasserman (2003) and introduces the general principles of QMC, drawbacks and advantages of using QMC, as well as our implementation of it.

The general objective in option pricing is to compute the expected discounted payoff. Under the risk-neutral measure, the option price can be written as

$$P = E^Q[f(U)] = \int_{[0,1]^s} f(x)dx \quad (9)$$

where $U = (U_1, \dots, U_s) \sim i.i.d. \text{ Unif}(0, 1)$. Standard MC approximates this integral by an equal-weight average over *i.i.d* drawn samples $U^{(i)}$:

$$\hat{P}_{MC} = \frac{1}{n} \sum_{i=1}^n f(U^{(i)}), \quad RMSE(\hat{P}_{MC}) = O(n^{-\frac{1}{2}}) \quad (10)$$

While MC is unbiased and supports simple confidence intervals, random points cluster

and leave gaps, which is exactly what drives the convergence rate $\frac{1}{\sqrt{n}}$. QMC replaces the *i.i.d.* draws with a deterministic low-discrepancy point set $x_1, \dots, x_n \in [0, 1]^s$ and uses the same equal-weight estimator:

$$\hat{P}_{QMC} = \frac{1}{n} \sum_{i=1}^n f(x_i) \quad (11)$$

Low discrepancy means these points are spread nearly uniformly across axis-aligned boxes in $[0, 1]^s$, where s is the dimension. More formally, a point set $P_n = \{x_1, \dots, x_n\} \subset [0, 1]^s$ has low-discrepancy if its star discrepancy

$$D_n^*(P_n) = \sup_{t \in [0, 1]^s} \left| \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{x_i \in [0, t]\} - \prod_{j=1}^s t_j \right| \quad (12)$$

satisfies $D_n^*(P_n) = O(\frac{(\log n)^s}{n})$. For integrands with bounded variation in the sense of Hardy-Krause, the Koksma-Hlawka inequality yields

$$\left| \frac{1}{n} \sum_{i=1}^n f(x_i) - \int_{[0, 1]^s} f(x) dx \right| \leq V_{HK}(f) D_n^*(P_n) \quad (13)$$

This result strictly bounds the integration error in equation (11). Hence, designing P_n to have small $D_n^*(P_n)$ turns geometric uniformity into a small worst-case quadrature error.

Although the Koksma-Hlawka inequality provides a worst-case error bound for QMC, it has limited practical value in option pricing. First, the bound requires f to have bounded variation in the sense of Hardy-Krause. Typically this regularity assumption is not satisfied in option pricing due to payoffs having discontinuities or kinks. Asian options violates this condition as the payoff is only piecewise smooth due to a kink along the surface where the average price equals the strike. Additionally, the discrepancy bound $D_n^*(P_n) = O(\frac{(\log n)^s}{n})$ depends on the dimension s , which in high dimensions drives the convergence rate towards the regular MC. For these reasons, we evaluate QMC performance empirically, in terms of the error decay of our estimators, rather than through the Koksma-Hlawka inequality (Moskowitz and Caffisch 1996).

4.1 Nets and sequences

The quasi-random points we consider are Sobol' sequences, which are constructed as digital (t, s) -sequences in base 2. Here t is the parameter describing uniformity of the sequence, s denotes the dimension $([0, 1]^s)$ and m controls the number of sample points. A (t, m, s) -net in base b is a finite set of points in $[0, 1]^s$ with $n = b^m$ points. Sobol' sequences are digital (t, s) -sequences in base $b = 2$, and for $n = 2^m$ their n points form (t, m, s) -nets. First we define the elementary interval in base b . The interval takes the

form:

$$\prod_{i=1}^s \left[\frac{a_i}{b^{j_i}}, \frac{a_i + 1}{b^{j_i}} \right]$$

with $j_i \in \{0, 1, \dots\}$ and $a_i \in \{0, 1, \dots, b^{j_i} - 1\}$. When $0 \leq t \leq m$, a (t, m, s) -net in base b is a set of $n = b^m$ points, such that, for every elementary interval of volume b^{t-m} , exactly b^t points fall in that interval. Thus, a smaller t results in greater uniformity. This is the so-called net property, which is crucial for the effectiveness of QMC. The net property enforces strong local uniformity and implies low discrepancy.

We use Sobol' sequences as they are commonly used for QMC in finance (Caflish, Morokoff, and A. Owen 1997; Glasserman 2003). Sobol' sequences work particularly well in our QMC setting as they are constructed as a (t, s) -sequence in base 2 for all dimensions s . This feature has computational advantages as base-2 digital nets are built from bitwise XOR's which allows points to be produced through cheap bit operations. Furthermore, the base-2 construction combines well with Brownian bridge construction, which reduces the effective dimension and results in even faster convergence (Glasserman (2003) and Moskowitz and Caflish (1996)). This matter will be elaborated in section 4.3. For a more in depth discussion of Sobol' sequences the reader is referred to Glasserman (2003, p.303-314).

4.2 Digital Scrambling

As QMC is purely deterministic, the estimator has no sampling distribution and therefore no sampling variance in the usual sense. This complicates the process of pricing an option with confidence intervals. By introducing a randomization step to QMC we combine the features of standard MC and QMC. Randomized QMC (RQMC) introduces randomness to retain low discrepancy on average while enabling standard error estimation across independent scramblings. The following is based on Art B. Owen (2019).

Formally, let $\mathcal{P}_n = \{x_1, \dots, x_n\} \subset [0, 1]^s$ be a deterministic low-discrepancy point set. An RQMC scheme constructs a randomized point set

$$\tilde{\mathcal{P}}_n(\omega) = \{\tilde{X}_1(\omega), \dots, \tilde{X}_n(\omega)\}$$

on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ such that each $\tilde{X}_i(\omega)$ is marginally $\text{Unif}([0, 1]^s)$, and conditional on ω , the point set $\tilde{\mathcal{P}}_n(\omega)$ retains the low-discrepancy structure of $\mathcal{P}_n$. The RQMC estimator based on a single randomization is

$$\hat{P}_{RQMC}^{(1)} = \frac{1}{n} \sum_{i=1}^n f(\tilde{X}_i),$$

which is now a random variable. Because each $\tilde{X}_i$ is uniformly distributed on $[0, 1)^s$, we have $\mathbb{E}[\hat{P}_{RQMC}^{(1)}] = P$ under the same regularity conditions that ensure $\mathbb{E}[f(U)] = P$ for $U \sim \text{Unif}([0, 1)^s)$. For digital nets and digital (t, d) -sequences such as Sobol', a natural randomization is digital scrambling in base 2 (Art B. Owen (1995, 1998)). Scrambling applies random permutations to the binary digits of each coordinate in such a way that each scrambled point $\tilde{X}_i$ remains $\text{Unif}([0, 1)^s)$, and for $n = 2^m$, the scrambled point set $\{\tilde{X}_1, \dots, \tilde{X}_n\}$ is still a (t, m, d) -net. Thus, scrambling preserves the net property and hence the low discrepancy, while introducing enough randomness to support standard error assessment.

In practice, we generate R independent scramblings of the same underlying Sobol' net. Let $\tilde{X}_i^{(r)}$ denote the i th point in the r th scrambled point set, for $r = 1, \dots, R$ and $i = 1, \dots, n$. Hence, each independent scrambling is generated by initializing the Sobol' generator with a different random seed, ensuring statistical independence between scramblings. The estimator from the r th scrambling is

$$\hat{P}^{(r)} = \frac{1}{n} \sum_{i=1}^n f(\tilde{X}_i^{(r)})$$

Our overall RQMC estimator is the average over scramblings ,

$$\hat{P}_{RQMC} = \frac{1}{R} \sum_{r=1}^R \hat{P}^{(r)}$$

Since each scrambling is constructed in the same way and the scramblings are independent, we have

$$\mathbb{E}[\hat{P}^{(r)}] = P \quad \text{for each } r, \quad \mathbb{E}[\hat{P}_{RQMC}] = P,$$

so the estimator is unbiased. Moreover, by independence across scramblings,

$$\text{Var}(\hat{P}_{RQMC}) = \text{Var}\left(\frac{1}{R} \sum_{r=1}^R \hat{P}^{(r)}\right) = \frac{1}{R^2} \sum_{r=1}^R \text{Var}(\hat{P}^{(r)}) = \frac{1}{R} \text{Var}(\hat{P}^{(1)})$$

In our numerical results we estimate $\text{Var}(\hat{P}^{(1)})$ by the sample variance of $\hat{P}^{(1)}, \dots, \hat{P}^{(R)}$:

$$\hat{\sigma}_R^2 = \frac{1}{R-1} \sum_{r=1}^R (\hat{P}^{(r)} - \hat{P}_{RQMC})^2,$$

So that an estimator of $\text{Var}(\hat{P}_{RQMC})$ is

$$\widehat{\text{Var}}(\hat{P}_{RQMC}) = \frac{\hat{\sigma}_R^2}{R}$$

This provides an empirical variance estimate across scramblings.

In our numerical setup we use the Sobol' generator from the `scipy.stats.qmc` module with `scramble=True` and draw $n = 2^m$ points. For each independent scrambling, we initialize the Sobol' generator with a different random seed to ensure independence between scramblings. This implementation applies a left linear matrix scrambling to the generating matrices of the Sobol' digital net, followed by a digital random shift in base 2 (Matoušek 1998). This combination scheme of linear matrix scrambling with a random shift is a digital scrambling of the type considered by Art B. Owen (1995, 1998). Here each scrambled point is uniformly distributed on $[0, 1]^s$, and for $n = 2^m$ the scrambled points remains a (t, m, s) -net.

4.3 Brownian Bridge Construction

In the RQMC setup, instead of constructing the Brownian motion incrementally as a random walk $W_{t_1}, \dots, W_{t_M}$ with independent increments, we generate the path using a Brownian bridge construction. The idea is to sample the endpoint of the Brownian motion first and then iteratively fill in intermediate points by conditioning on its neighbours. This reordering concentrates most of the path variability in the first few driving random variables, which is beneficial for RQMC due to its first dimensions being very uniformly distributed (Glasserman 2003; Moskowitz and Caffisch 1996).

Let $0 = t_0 < t_1 < \dots < t_M = T$ be a fixed time grid. A standard discretization of Brownian motion would simulate

$$W_{t_i} = W_{t_{i-1}} + \sqrt{t_i - t_{i-1}} Z_i, \quad i = 1, \dots, M,$$

with $Z_1, \dots, Z_M \sim \text{i.i.d. } N(0, 1)$. In the Brownian bridge construction we instead proceed by first generating the endpoint $W_{t_M} \sim N(0, T)$. Then given two already simulated values W_{t_i} and W_{t_j} with $t_i < t_k < t_j$, the conditional distribution of W_{t_k} is

$$W_{t_k} \mid W_{t_i}, W_{t_j} \sim N\left(\frac{t_j - t_k}{t_j - t_i} W_{t_i} + \frac{t_k - t_i}{t_j - t_i} W_{t_j}, \frac{(t_k - t_i)(t_j - t_k)}{t_j - t_i}\right) \quad (14)$$

We use this formula to simulate midpoints between known grid points. Starting from $W_{t_0} = 0$ and the endpoint W_{t_M} , we first draw the midpoint $W_{t_{M/2}}$, then the midpoints of the two subintervals, and so on, until all $W_{t_1}, \dots, W_{t_{M-1}}$ have been generated. In practice, the Brownian bridge can be written as a linear mapping

$$(W_{t_1}, \dots, W_{t_M})^\top = BZ,$$

where $Z = (Z_1, \dots, Z_M)^\top$ are i.i.d. $N(0, 1)$ variables obtained from the Sobol' points via the inverse normal transform, and B is a deterministic lower-triangular matrix encoding the bridge construction. The difference from the standard random walk is the ordering in

which the coordinates of Z enter the path. Here early coordinates determine the endpoint and large-scale features of the path, while later coordinates only refine local fluctuations.

In our implementation, each scrambled Sobol' point $u = (u_1, \dots, u_M) \in [0, 1]^M$ is first mapped to a Gaussian vector $Y_i = \Phi^{-1}(u_i)$ and then to the Brownian path $(W_{t_i})_{i=1}^M$ via the Brownian bridge. The increments $\Delta W_i = W_{t_i} - W_{t_{i-1}}$ are then used in the two discretization schemes from Section 3 to simulate the CEV dynamics. This keeps the discretization of the CEV process unchanged and only the construction of the driving Brownian motion differs. As Moskowitz and Caflisch (1996) and Caflisch, Morokoff, and A. Owen (1997) argue, the implementation of Brownian bridge construction reduces the effective dimension to increase the effectiveness of RQMC. The algorithm for Brownian bridge construction can be found in appendix A.2.

5 Variance Reduction Techniques

The Monte Carlo simulation is a flexible and unbiased method for pricing options under complex dynamics such as the Constant Elasticity of Variance (CEV) model. However, it often suffers from high variance and slow $O(n^{-1/2})$ convergence. When pricing arithmetic Asian options under the CEV model, the payoff depends on the entire simulated price path. This path dependence increases the variance of the estimator, making variance reduction important.

To obtain accurate estimates of prices and sensitivities, we use variance reduction techniques to improve efficiency without introducing bias. In this paper we apply control variates and antithetic variates. Control variates reduce variance by adjusting estimates using correlated quantities with known expectations, while antithetic variates reduce variance by introducing negative dependence between simulated paths.

The theoretical framework in this section closely follows the presentation and notation of Glasserman (2003).

5.1 Control Variates

The idea of the control variate method is to use an additional variable X_i that is correlated with the simulated payoff Y_i and has a known expected value $\mathbb{E}[X]$. The method adjusts the simulated payoffs Y_i according to how its corresponding X_i differs from its expected value. In this way the variance of the estimator can be reduced without changing its mean.

We let $Y_1, \dots, Y_N$ be the simulated payoffs from N independent paths, where each Y_i represents the discounted option payoff obtained in the i th simulation. The goal is to estimate the expected value $\mathbb{E}[Y]$ using these samples. To apply the control variate

method, we need a few assumptions to hold. Firstly we need the pairs (X_i, Y_i) across simulations to be i.i.d. and have finite first and second moments (i.e. a well defined mean and variance). Furthermore we require the mean of the control variates $\mathbb{E}[X]$ to be known, otherwise we could introduce bias into the estimator. We also need X and Y to be linearly correlated, i.e. $\text{Cov}(Y, X) \neq 0$, in order for variance reduction to actually occur. With these conditions the control variate estimator remains unbiased and consistent.

The standard Monte Carlo estimator is given by:

$$\bar{Y} = \frac{1}{N} \sum_{i=1}^N Y_i \quad (15)$$

To reduce the variance of the estimator $\bar{Y}$, we introduce a control variate adjustment. For any constant $b \in \mathbb{R}$:

$$Y_i(b) = Y_i - b(X_i - \mathbb{E}[X]) \quad (16)$$

When we average over all replications, we get the control variate estimator:

$$\bar{Y}(b) = \frac{1}{N} \sum_{i=1}^N Y_i(b) = \bar{Y} - b(\bar{X} - \mathbb{E}[X]) \quad (17)$$

Since $\mathbb{E}[\bar{X}] = \mathbb{E}[X]$, the estimator $\bar{Y}(b)$ is unbiased for any value of b . The parameter b determines the amount of variance reduction and will be chosen to minimize $\text{Var}(Y_i(b))$. The variance of $Y_i(b)$ can be expressed as:

$$\begin{aligned} \text{Var}[Y_i(b)] &= \text{Var}[Y - b(X - \mathbb{E}[X])] \\ &= \text{Var}[Y] - 2b \text{Cov}(X, Y) + b^2 \text{Var}[X]. \end{aligned} \quad (18)$$

Minimizing this quadratic expression with respect to b gives us:

$$b^* = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} \quad (19)$$

Now we substitute b^* into the variance expression from equation (18) and get:

$$\begin{aligned}
 \text{Var}[Y_i(b^*)] &= \text{Var}[Y] - 2 \frac{\text{Cov}(X, Y)}{\text{Var}[X]} \text{Cov}(X, Y) + \left(\frac{\text{Cov}(X, Y)}{\text{Var}[X]} \right)^2 \text{Var}[X] \\
 &= \text{Var}[Y] - 2 \frac{\text{Cov}(X, Y)^2}{\text{Var}[X]} + \frac{\text{Cov}(X, Y)^2}{\text{Var}[X]^2} \text{Var}[X] \\
 &= \text{Var}[Y] - 2 \frac{\text{Cov}(X, Y)^2}{\text{Var}[X]} + \frac{\text{Cov}(X, Y)^2}{\text{Var}[X]} \\
 &= \text{Var}[Y] - \frac{\text{Cov}(X, Y)^2}{\text{Var}[X]}
 \end{aligned} \tag{20}$$

We recall that the correlation coefficient is defined as:

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}[X] \text{Var}[Y]}}, \tag{21}$$

which implies that:

$$\rho_{XY}^2 = \frac{\text{Cov}(X, Y)^2}{\text{Var}[X] \text{Var}[Y]} \Leftrightarrow \text{Cov}(X, Y)^2 = \rho_{XY}^2 \text{Var}[X] \text{Var}[Y] \tag{22}$$

Substituting this into equation gives us:

$$\begin{aligned}
 \text{Var}[Y_i(b^*)] &= \text{Var}[Y] - \frac{\text{Cov}(X, Y)^2}{\text{Var}[X]} \\
 &= \text{Var}[Y] - \frac{\rho_{XY}^2 \text{Var}[X] \text{Var}[Y]}{\text{Var}[X]} \\
 &= \text{Var}[Y] - \rho_{XY}^2 \text{Var}[Y] \\
 &= (1 - \rho_{XY}^2) \text{Var}[Y]
 \end{aligned} \tag{23}$$

The variance reduction is determined by the strength of the correlation between X and Y . As $|\rho_{XY}|$ gets closer to one, the variance reduction increases, and in the ideal case $|\rho_{XY}| = 1$ the variance becomes zero.

In practice, the optimal coefficient b^* depends on the unknown covariance term $\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$ and the unknown variance term $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$. These are population quantities, meaning that they describe the true underlying distribution. Since these are unknown, they are estimated from the simulated samples. The sample estimate of b^* is given by:

$$\hat{b}_N = \frac{\sum_{i=1}^N (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^N (X_i - \bar{X})^2} \tag{24}$$

For simplicity, we will refer to this estimate as $\hat{b}$ in the following. Using this coefficient,

the control variates estimator is defined as:

$$\bar{Y}_{CV} = \bar{Y}(\hat{b}) = \bar{Y} - \hat{b}(\bar{X} - \mathbb{E}[X]) \quad (25)$$

where $\bar{Y}_{CV}$ denotes the control variates estimator of $\mathbb{E}[Y]$. The estimated coefficient $\hat{b}$ corresponds to the slope from a least-squares regression between the simulated values of X_i and Y_i . As the number of simulations N increases, $\hat{b}_N$ converges to the optimal value b^* by the law of large numbers. Using $\hat{b}$ instead of the true b^* makes the adjusted values $Y_i(\hat{b})$ slightly dependent. This can introduce a small bias when N is finite, however since the bias is of order $1/N$ the bias becomes negligible as the number of simulations grows.

5.2 Antithetic Variates

In the following section we focus on the variance reduction technique called antithetic variates. This method reduces variance by introducing negative correlation between simulated paths instead of regression adjustment as in the control variates method. Lets say that we run a Monte Carlo simulation with random draws Z from a standard normal distribution $N(0, 1)$. For each draw we create a mirrored value $-Z$, in order to get negatively correlated outputs Y_i and $\tilde{Y}_i$. If one simulated payoff happens to be unusually high, its antithetic pair tends to be low, so their average stabilizes the estimate and smooths out random noise in the simulation.

We let $Y = f(U)$ denote the simulated output, where $U = (U_1, \dots, U_M)$ is a vector of i.i.d. random variables with a uniform distribution on $[0, 1]$. We create an antithetic path using $1 - U = (1 - U_1, \dots, 1 - U_M)$ which has a payoff given by $\tilde{Y}_i = f(1 - U)$. Y_i and $\tilde{Y}_i$ has the same distribution, but are negatively correlated through their dependence on U . We estimate $\mathbb{E}[Y]$ using N simulated pairs $(Y_i, \tilde{Y}_i)$ for $i = 1, \dots, N$. To apply the antithetic variates method, we make the following assumptions: Y and $\tilde{Y}$ need to have finite first and second order moments. $(Y_i, \tilde{Y}_i)$ are i.i.d. across i , while Y_i and $\tilde{Y}_i$ within each pair are dependent through shared random inputs U_i . Each U_i consists of i.i.d. Uniform(0, 1) components, and the antithetic pair $(U_i, 1 - U_i)$ ensures that Y_i and $\tilde{Y}_i$ are identically distributed but negatively correlated. Lastly, both Y_i and $\tilde{Y}_i$ are unbiased for $\mathbb{E}[Y]$, so the estimator $\hat{Y}_{AV}$ remains unbiased.

Several of these assumptions are similar to those used in the control variates method, however in the antithetic variates method we generate negatively correlated payoffs through mirrored random inputs instead of using a correlated control variable with a known expectation.

The antithetic variates estimator is defined as:

$$\hat{Y}_{AV} = \frac{1}{N} \sum_{i=1}^N \left(\frac{Y_i + \tilde{Y}_i}{2} \right) \quad (26)$$

This estimator averages each antithetic pair and then takes the mean across all pairs. Since both Y_i and $\tilde{Y}_i$ are unbiased for $\mathbb{E}[Y]$, their average is also an unbiased estimator of $\mathbb{E}[Y]$. This ensures that the variance is reduced without introducing bias into the estimator. Now the variance of the antithetic estimator is derived:

$$\begin{aligned} \text{Var}[\hat{Y}_{AV}] &= \text{Var} \left[\frac{1}{N} \sum_{i=1}^N \left(\frac{Y_i + \tilde{Y}_i}{2} \right) \right] \\ &= \frac{1}{N} \text{Var} \left[\frac{Y + \tilde{Y}}{2} \right] \\ &= \frac{1}{N} \left(\frac{1}{2} \right)^2 \text{Var}[Y + \tilde{Y}] \\ &= \frac{1}{4N} (\text{Var}[Y] + \text{Var}[\tilde{Y}] + 2\text{Cov}(Y, \tilde{Y})) \end{aligned} \quad (27)$$

Since $\text{Var}[Y] = \text{Var}[\tilde{Y}]$ we can simplify the expression to:

$$\text{Var}[\hat{Y}_{AV}] = \frac{1}{2N} \text{Var}[Y] (1 + \rho_{Y\tilde{Y}}) \quad (28)$$

, where $\rho_{Y\tilde{Y}}$ is the correlation between Y and $\tilde{Y}$. When $\rho_{Y\tilde{Y}} < 0$, the variance decreases, and the more negative this correlation is, the greater the decrease is. If $\rho_{Y\tilde{Y}} = -1$ the variance becomes 0.

In practice, the effectiveness of the variance reduction method depends on the payoff being monotonic in the random inputs. When the payoff function $f(U)$ is monotonic, it means that the payoff always moves in one direction, i.e. as the input U increases, the payoff either increases or decreases, but not both. Hence large values of U lead to large values of $Y = f(U)$ and small values of U lead to small values of $\tilde{Y} = f(1 - U)$. This creates a negative correlation between the paired payoffs $(Y_i, \tilde{Y}_i)$, resulting in a smoother estimator with reduced variance.

6 Monte Carlo Estimation of Delta and Vega

In this section, we present how we estimate the sensitivities (or Greeks) of the arithmetic Asian option price with respect to key model parameters under the CEV model. The sensitivities quantify how the option value changes when the underlying parameters are perturbed. In particular, we compute Delta and Vega, which measure the sensitivity of the option price to changes in the initial asset price S_0 and volatility σ , respectively. Since closed-form solutions for these Greeks are unavailable under the CEV model, we employ the finite-difference (FD) approach, as described in Glasserman (2003). Several alternative Monte Carlo estimators for Greeks exist, including pathwise derivative and likelihood ratio methods. In this study, the finite difference approach was chosen as a straightforward and robust benchmark.

Let $P(S_0, \sigma)$ denote the true price of the Asian option under the risk neutral measure, then we can represent Delta and Vega as

$$\text{Delta} = \frac{\partial P}{\partial S_0}, \quad \text{Vega} = \frac{\partial P}{\partial \sigma}$$

Assuming that P is differentiable in $\theta = \{S_0, \sigma\}$. Note, that both sensitivities reflect the effect of parameter changes over the entire path, not just the terminal value.

To find these derivatives, we use the indirect simulation based method FD. If $P(\theta)$ denotes the estimated price of underlying asset which depends on a chosen parameter, θ (either S_0 or σ), then we can write up the *forward-difference* estimator as

$$\hat{\Delta}_F = \frac{P(\theta + \epsilon) - P(\theta)}{\epsilon}$$

where ϵ is some small perturbation. This estimator is simple and intuitive but biased to first order in ϵ . To reduce bias, we instead use the central-difference estimator, where we simulate at both $\theta - \epsilon$ and $\theta + \epsilon$:

$$\hat{\Delta}_C = \frac{P(\theta + \epsilon) - P(\theta - \epsilon)}{2\epsilon}$$

The central-difference estimator improves the rate of convergence of the bias term from $\mathcal{O}(\epsilon)$ to $\mathcal{O}(\epsilon^2)$, although it requires an additional simulation run compared to the forward difference method. We use the central-difference estimator due to its better rate of convergence, but for a more thorough discussion the reader is referred to Glasserman (2003).

The finite difference step size ϵ is chosen as a fixed relative perturbation to the baseline parameter value. Smaller values of ϵ reduce truncation bias but amplify Monte Carlo noise through subtraction error, whereas larger ϵ increase bias. The choice of ϵ reflects a

practical bias–variance trade-off which needs to be carefully chosen

When estimating we will use common random numbers for the different ϵ , such that the simulations are using the same underlying draws, reducing variance.

7 Numerical Results

7.1 Experimental Setup

We need to choose the level of both volatility, σ and elasticity of the volatility, γ for our CEV model. A number of empirical studies have estimated CEV parameters for different commodities under the physical probability measure. For example, Čermák (2017) estimates CEV models for corn and wheat using daily spot prices from 2010–2014 and reports elasticity parameters of approximately $\gamma = 1.26$ (corn) and $\gamma = 1.21$ (wheat).

An older study finds a wide range of elasticity levels across commodity classes. Geman and Shih (2009) found that crude oil typically exhibited $\gamma = 1.05$ (1990-1999) and around $\gamma = 0.68$ (2000-2007), while metals such as copper and gold often yield $\gamma = 1.1 - 1.5$.

Since these estimates are obtained under the real-world measure and reflect historical return variability rather than option-implied volatilities, we do not use their reported volatility coefficients directly for pricing. Instead, we use these studies as motivation for choosing plausible ranges for γ , fix the drift to the risk-free rate and use the same σ and risk-free rate as Hickernell, Lemieux, and Art B Owen (2005), who also investigated the Asian call option and the effect of control variates using QMC.

All simulations were performed using fixed model parameters, identical time grids for all estimators, and consistent random number generation across variance reduction techniques where applicable. Runtime measurements reflect both algorithmic efficiency and implementation-level effects such as vectorization. While exact wall-clock time may vary across hardware platforms, relative performance comparisons between methods remain qualitatively stable.

In our numerical analysis, we focus on At-The-Money(ATM) Asian call options, where $K = S_0$. This choice is made because ATM options provide a balanced payoff structure. They are neither almost always worthless, as deep out-of-the-money options, nor almost always exercised, as deep in-the-money options. As a result, ATM options exhibit meaningful payoff variability, making them well suited for comparing Monte Carlo methods, variance reduction techniques, and Greek estimators.

Lastly, the algorithm for pricing Asian options with scrambling can be found in appendix A.3.

7.2 Euler-Maruyama and Milstein Discretization

To assess how the choice of time discretization affects the accuracy of the simulated CEV process, we compare the mean-square error (MSE) of the Euler-Maruyama and Milstein schemes as the time-step $\Delta t = T/M$ decreases. For each Δt , the MSE at maturity T is computed relative to a high-resolution benchmark generated using a very fine Milstein grid ($M=8.192$). The results are displayed in Figure 1. The empirical log-log slopes align closely with the theoretical strong convergence rates: approximately 0.5 for Euler-Maruyama and 1.0 for Milstein. This demonstrates that Milstein delivers significantly higher accuracy for a given step size, as expected when the diffusion coefficient is differentiable. Although Milstein carries a slightly higher per-step computational cost due to the additional derivative term, its improved accuracy seem to compensate for this. Because the remainder of our numerical experiments rely on precise path simulation, we adopt the Milstein scheme for all subsequent simulations.

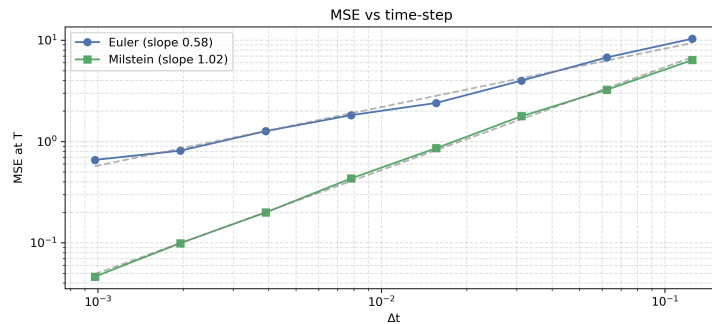


Figure 1: Strong convergence of Euler-Maruyama and Milstein
 Estimated with $N = 2^{16} = 65.536$, $S_0 = 100$, $K = 100$, $T = 1$, $\sigma = 0.3$, $\gamma = 1.2$

7.3 Crude Monte Carlo vs. Randomized Quasi-Monte Carlo

7.3.1 Effect of the Brownian Bridge on RQMC Performance

As discussed in section 4.3 we use the Brownian bridge construction to reduce effective dimension when doing RQMC. In this section we will briefly compare the RQMC with and without the Brownian bridge construction.

In Table 1, the VRF have been calculated as $(\text{BB SE}/\text{plain SE})^2$ and Time ratios have been calculated by taking the Brownian bridge runtime and dividing by the plain RQMC's time. As we see can see there is a decrease in the variance of around 5-60x, while the

running time increases by around 6%-27%. This trade-off implies a considerable net improvement in efficiency. Under the usual $\frac{1}{\sqrt{N}}$ scaling, achieving such reductions with standard RQMC alone would require increasing the number of simulated paths by approximately 10 to 60 times.

N	Price _{RQMC}	SE _{RQMC}	Price _{RQMCBB}	SE _{RQMCBB}	VRF	Time ratio
256	17.8158	0.2703	17.8806	0.1192	5.1x	1.06
512	17.9367	0.1538	17.8535	0.0531	8.4x	1.17
1,024	17.8035	0.0955	17.8619	0.0276	11.9x	1.11
2,048	17.8530	0.0545	17.8293	0.0180	9.2x	1.12
4,096	17.8694	0.0389	17.8332	0.0087	20.2x	1.14
8,192	17.8589	0.0262	17.8378	0.0087	9.1x	1.11
16,384	17.8350	0.0171	17.8491	0.0076	5.0x	1.17
32,768	17.8433	0.0144	17.8489	0.0036	15.5x	1.19
65,536	17.8528	0.0110	17.8486	0.0019	33.5x	1.27
131,072	17.8468	0.0065	17.8461	0.0010	41.7x	1.26
262,144	17.8483	0.0054	17.8458	0.0007	59.9x	1.21

Table 1: RQMC with and Without Brownian Bridge

Estimated with $S_0 = 100$, $K = 100$, $T = 1$, $\sigma = 0.3$, $\gamma = 1.2$ and 32 scrambles

As seen in Figure 2, by plotting the efficiency (VRF/Time ratio) to account for both precision and computational cost. Values are strictly above 1 across all sample sizes, which indicates that the introduction of the Brownian Bridge is consistently more efficient than the standard RQMC. In particular, for larger simulation sizes, the efficiency metric seem to grow even larger. The Brownian bridge even achieves 50 times the efficiency of the unmodified RQMC at $N = 262,144$. This suggests that the Brownian Bridge effectively reduces the effective dimension of the underlying stochastic process by allocating the early Sobol coordinates to the most influential time steps of the Brownian path. Consequently, the estimator converges faster and exhibits much smaller sampling variability at nearly the same computational cost.

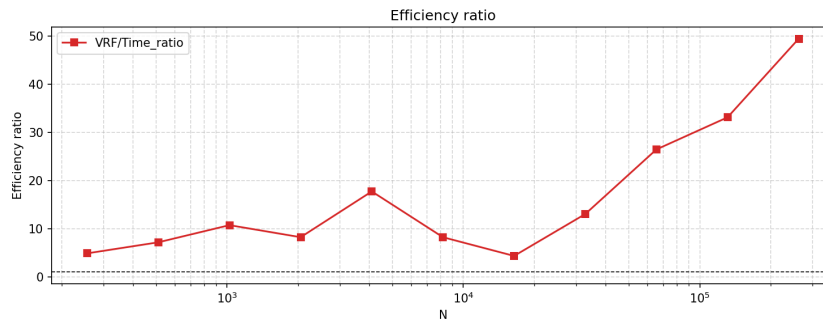


Figure 2: Relative Efficiency of using Brownian Bridge

Estimated with $S_0 = 100$, $K = 100$, $T = 1$, $\sigma = 0.3$, $\gamma = 1.2$ and 32 scrambles.

Brownian Bridge is preferred if the efficiency is above 1

In summary, our results provide strong empirical support for using the Brownian Bridge in quasi-Monte Carlo simulations. Because the Brownian bridge is just a deterministic linear transformation of the underlying Gaussian vector, it leaves the expectation, and hence the unbiasedness, of the Monte Carlo estimator unchanged. Empirically, we observe that it drastically decreases variance and achieves significant gains in overall computational efficiency, making it a natural and powerful enhancement to RQMC methods for path-dependent option pricing under the CEV model, which is why we will use the Brownian bridge, going further in this paper.

7.3.2 Convergence and Accuracy of CMC and RQMC

As we have established that Milstein is our preferred discretization scheme, we now turn to a comparison of simulation methods. In particular, we compare the convergence behaviour and accuracy of the Crude Monte Carlo (CMC) estimator with those of a Randomized Quasi-Monte Carlo (RQMC) estimator implemented using a Brownian bridge construction. Both methods are used to price the arithmetic Asian call option under the CEV model using identical parameters. This allows us to isolate the effect of the sampling method.

As illustrated in Figure 3, both CMC and RQMC converge toward the same mean option price as the number of simulated paths (N) increases. We observe that the confidence intervals (CIs) are much tighter for RQMC across all N than for CMC. However, it is important to note that RQMC's CI reflect only variability across independent digital scramblings of the Sobol' sequence rather than classical Monte Carlo sampling variability. Furthermore, CMC shows noticeable fluctuations for smaller N , while RQMC remains stable even at low N . Hence from the figure we see that RQMC has a faster convergence and lower estimator variance.

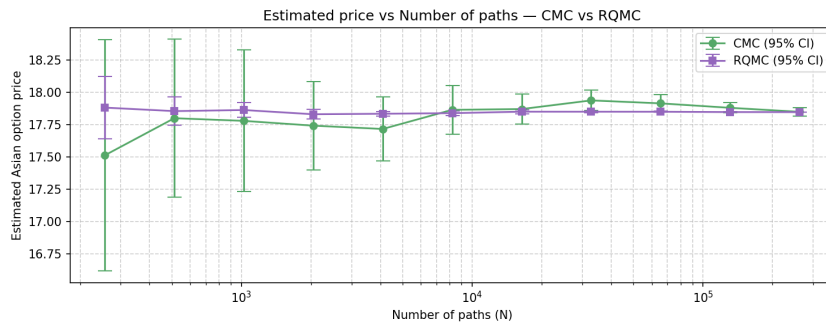


Figure 3: Estimated Asian option price as a function of the number of simulated paths (N), showing the mean price estimates and corresponding 95% confidence intervals obtained from the CMC and RQMC simulations.

Estimated with $S_0 = 100$, $K = 100$, $\sigma = 0.3$, $\gamma = 1.2$ and 32 scrambles.

From Figure 4 we observe that as we increase the number of simulated paths (N), the standard error (SE) decreases. We notice that the RQMC graph consistently has a lower SE than CMC for all N . This indicates that RQMC achieves a higher precision for the same number of simulations.

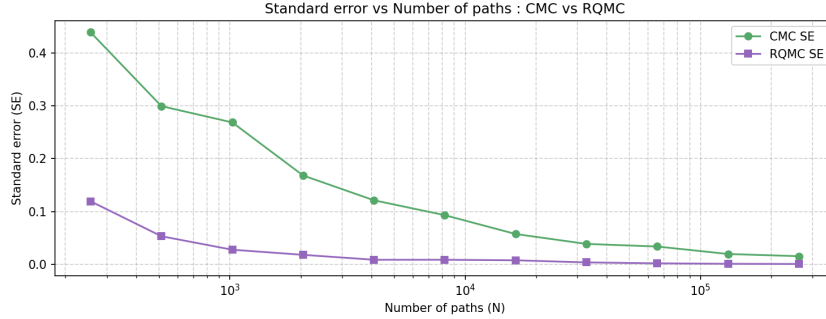


Figure 4: Convergence of standard errors for CMC and RQMC
Estimated with $S_0 = 100$, $K = 100$, $\sigma = 0.3$, $\gamma = 1.2$ and 32 scrambles.

Table 2 further shows this difference. For example, the standard error for CMC is $SE_{mc} = 0.0196$ at $N = 131,072$ with a runtime of ≈ 84 seconds, whereas RQMC achieves almost the same precision with $SE_{qmc} = 0.0180$, but at $N = 2,048$ with a runtime of ≈ 2 seconds. This shows that RQMC attains the same level of accuracy at a much lower computational cost than CMC. Furthermore we observe that as N increases, the variance reduction factor (VRF) also increases - from $13.6\times$ at $N = 256$ to $496.3\times$ at $N = 262,144$. This means that the variance of the RQMC estimator is roughly 14 times smaller than that of CMC at the smallest N , and almost 500 times smaller at the largest N . Thus, the variance reduction achieved by RQMC becomes stronger for larger simulations, and RQMC produces lower standard errors than CMC across all reported values of N . However, this is conditional on the parameter values and discretization used in this study.

In summary, both methods converge to the same expected option price, but RQMC achieves substantially lower variance. Its advantage increases with the number of simulations, which allows RQMC to reach a given precision with fewer simulations than CMC.

7.4 Performance of Variance Reduction Techniques

7.4.1 Effect of Control Variates

To assess the effectiveness of the control variate method within CMC, we used the geometric Asian option (under GBM) as a control variate for the arithmetic Asian option under the CEV model. The geometric Asian option is chosen because its payoff is strongly related to that of the arithmetic Asian option, yet its expected value is known in closed form. Although our underlying follows the CEV dynamics, both arithmetic and geometric averages are driven by the same simulated price paths, and the geometric average remains a smooth, monotone transformation of the arithmetic one. We emphasize that the expected

N	Price _{mc}	SE _{mc}	Time _{mc}	Price _{qmc}	SE _{qmc}	Time _{qmc}	VRF
256	17.5120	0.4388	0.6813	17.8806	0.1192	1.2160	13.6x
512	17.7987	0.2992	0.7784	17.8535	0.0531	1.3368	31.7x
1,024	17.7780	0.2684	0.9948	17.8619	0.0276	1.7051	94.3x
2,048	17.7410	0.1681	1.4449	17.8293	0.0180	2.2180	87.4x
4,096	17.7154	0.1212	2.2258	17.8332	0.0087	3.5735	196.1x
8,192	17.8635	0.0930	3.9704	17.8378	0.0087	6.2023	115.6x
16,384	17.8694	0.0574	7.2444	17.8491	0.0076	11.7064	56.4x
32,768	17.9363	0.0386	14.6825	17.8489	0.0036	23.5221	112.0x
65,536	17.9134	0.0337	32.5369	17.8486	0.0019	55.2950	313.2x
131,072	17.8790	0.0196	83.9481	17.8461	0.0010	183.1927	376.1x
262,144	17.8480	0.0154	200.0175	17.8458	0.0007	416.1157	496.3x

Table 2: Summary statistics for CMC and RQMC simulations. The table includes the estimated option prices, standard errors, runtimes (s), and VRFs ($\frac{SE_{mc}^2}{SE_{qmc}^2}$) for different numbers of simulated paths N . Estimated with $S_0 = 100$, $K = 100$, $\sigma = 0.3$, $\gamma = 1.2$ and 32 scrambles.

tation of the geometric Asian option used in the control adjustment is taken under the Black–Scholes (GBM) model rather than under the CEV dynamics. The control variate is therefore not derived from an exact model-consistent identity, but is used as an approximate control whose effectiveness relies purely on empirical correlation rather than exact analytical equivalence. This structural similarity produces a high correlation between the two payoffs—especially when γ is close to 1, where the CEV model approaches GBM. We would therefore expect this to be an effective control variate. Furthermore the availability of the analytical price ensures that the adjustment does not introduce bias while reducing variance in the estimator.

In Table 3, we ran the simulation multiple times with and without control variates, with different values for γ (also illustrated in Figure 5). We see that the correlation between the control and target payoffs increases when gamma is close to 1, reaching correlations close to 1. The Variance Reduction Factor (VRF) grows with this correlation and peaks around $\gamma = 0.98$, which improved the precision by just 940 times relative to the CMC estimator by only increasing the running time by 5%. This is however not very useful, as such a high correlation suggests that the geometric Asian option already serves as a very accurate proxy for the arithmetic Asian option in this parameter region.

If we instead look at $\gamma = 1.2$, which we chose for our numerical setup, we see a bit more modest VRF of 6.9, only increasing the running time by 6%. The increment of the running time is fairly constant, extending the computation time by 0 – 9%. For our setup with $\gamma = 1.2$, there is still a correlation of 0.92 with the control variate which makes it still effective for reducing variance.

Figure 5, plots VRF by the different γ 's. The VRF peaks when γ is close to 1, because correlation is high. The "Bell" of the VRF's, also do not seem to be perfectly symmetrical, and the control variate actually seem to favour the leverage effect ($\gamma < 1$), suggesting that the control variate may perform better when modelling the underlying as an equity rather than a commodity. However, we could say that, the effect of the control variate on the CMC estimator, heavily depends on what the underlying asset is and how it is modelled.

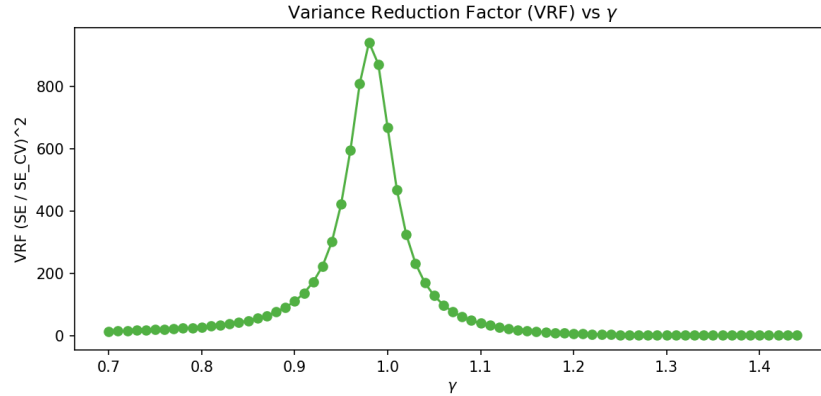


Figure 5: Control Variate CMC: Variance Reduction Factor and Time ratio plotted for varying γ for CMC.

Estimated with $S_0 = 100$, $K = 100$, $T = 1$, $\sigma = 0.3$, $N = 16,384$ and 32 scrambles.

γ	Corr	SE CMC	SE CMC+CV	VRF	Time ratio
0.50	0.8840	0.0020	0.0009	5.5x	1.00
0.60	0.9257	0.0030	0.0011	8.1x	1.06
0.70	0.9603	0.0044	0.0012	14.1x	1.09
0.80	0.9826	0.0066	0.0012	28.0x	1.06
0.90	0.9951	0.0101	0.0010	109.9x	1.05
0.95	0.9983	0.0127	0.0006	422.0x	1.06
0.96	0.9986	0.0134	0.0005	595.2x	1.05
0.97	0.9989	0.0140	0.0005	808.3x	1.08
0.98	0.9991	0.0147	0.0005	940.9x	1.05
0.99	0.9992	0.0154	0.0005	870.5x	1.06
1.00	0.9991	0.0162	0.0006	667.4x	1.05
1.01	0.9989	0.0171	0.0008	467.5x	1.07
1.02	0.9986	0.0179	0.0010	324.3x	1.05
1.03	0.9980	0.0189	0.0012	231.7x	1.07
1.04	0.9974	0.0199	0.0015	169.6x	1.06
1.05	0.9965	0.0210	0.0019	128.1x	1.07
1.10	0.9879	0.0280	0.0044	39.8x	1.07
1.20	0.9211	0.0574	0.0218	6.9x	1.06
1.30	0.6685	0.2025	0.1887	1.2x	1.05
1.40	0.2934	0.3640	0.3314	1.2x	1.06
1.50	0.0681	1.1956	1.2037	1.0x	1.08

Table 3: Control variates CMC: Table of VRF and time ratios

Estimated with $S_0 = 100$, $K = 100$, $T = 1$, $\sigma = 0.3$, $N = 16,384$ and 32 scrambles.

If we apply the control variate to RQMC, the picture changes noticeably. Table 4, reports the standard errors, variance reduction factors and time ratios for different values of γ . Keeping the sample size the same as for MC, $N = 16,384$, RQMC without CV already

yields small standard errors, and adding the control variate, gives a more modest VRF compared to CMC. For a low $\gamma = 0.7$ the CV actually increase the standard errors as the $VRF = 0.5$. Only when γ is closer to 1 do we obtain substantial VRFs, but even then the gain is partly offset by the higher computational cost, with a runtime increase of around 13–31%.

For CMC, the same control variate produces greater variance reductions when γ is near 1 at essentially no extra cost, because the estimator is still dominated by sampling noise and a highly correlated control variate can remove most of that noise. However, in RQMC, scrambled Sobol' points combined with a Brownian bridge have already removed a lot of the variance by improving the space-filling properties and concentrating most of the path variability in the first dimensions. There is not much variance left for the CV to remove so the incremental effect of CV is smaller. In Table 4, we have decreased number of paths to get closer to the same standard error as CMC. We actually see that we can get a higher VRF for different γ 's, but this is not consistent with all γ 's, as for some they decrease the VRF. On thing that is consistent is the increase in the time ratios. This is because the estimation of $\hat{b}$ becomes relatively more expensive as the run time of the RQMC decreases. This is illustrated when $N = 512$, where it actually takes more than twice the time to include the control variate.

γ	Corr	SE RQMC	SE RQMC+CV	VRF	Time ratio
0.70	0.9595	0.0005	0.0008	0.5x	1.31
0.80	0.9823	0.0010	0.0008	1.7x	1.20
0.90	0.9950	0.0016	0.0006	6.8x	1.21
0.95	0.9983	0.0021	0.0004	21.0x	1.21
0.96	0.9987	0.0022	0.0004	27.1x	1.19
0.97	0.9989	0.0023	0.0004	34.3x	1.13
0.98	0.9991	0.0024	0.0004	41.1x	1.19
0.99	0.9992	0.0025	0.0004	43.4x	1.27
1.00	0.9992	0.0027	0.0004	39.9x	1.19
1.01	0.9990	0.0028	0.0005	32.2x	1.20
1.02	0.9986	0.0030	0.0006	23.9x	1.20
1.03	0.9981	0.0032	0.0008	17.5x	1.20
1.04	0.9975	0.0034	0.0009	13.1x	1.13
1.05	0.9966	0.0036	0.0011	10.0x	1.20
1.10	0.9879	0.0053	0.0028	3.5x	1.19
1.20	0.9086	0.0171	0.0159	1.2x	1.20
1.30	0.3888	0.2067	0.2049	1.0x	1.20
1.40	0.2059	0.3061	0.3071	1.0x	1.20

γ	Corr	SE RQMC	SE RQMC+CV	VRF	Time ratio
0.70	0.9590	0.0012	0.0018	0.4x	2.13
0.80	0.9823	0.0024	0.0022	1.2x	2.13
0.90	0.9950	0.0042	0.0018	5.6x	2.11
0.95	0.9982	0.0055	0.0014	16.3x	2.14
0.96	0.9986	0.0058	0.0013	21.2x	2.13
0.97	0.9989	0.0061	0.0011	28.4x	2.14
0.98	0.9990	0.0064	0.0011	36.9x	2.27
0.99	0.9991	0.0067	0.0010	44.6x	2.10
1.00	0.9991	0.0071	0.0010	48.7x	2.10
1.01	0.9989	0.0074	0.0011	44.3x	2.11
1.02	0.9987	0.0078	0.0014	32.9x	2.14
1.03	0.9982	0.0082	0.0017	22.7x	2.11
1.04	0.9976	0.0086	0.0022	15.6x	2.14
1.05	0.9968	0.0091	0.0028	10.6x	2.13
1.10	0.9895	0.0132	0.0077	2.9x	2.14
1.20	0.9412	0.0531	0.0479	1.2x	2.13
1.30	0.8071	0.2851	0.2763	1.1x	2.11
1.40	0.6957	2.4627	2.4522	1.0x	2.16

a) N=16.384

b) N=512

Table 4: Control variates RQMC: Table of VRF and time ratios

Estimated with $S_0 = 100$, $K = 100$, $T = 1$, $\sigma = 0.3$ and 32 scrambles.

Hickernell, Lemieux, and Art B Owen (2005) show that one should expect CMC to have a bigger gain than QMC when using CV. They found that MC and QMC generally helped with reducing variance, but the gains from using CV are typically smaller for QMC than for CMC. They also emphasize that the optimal CV coefficient $\hat{b}$, for QMC is not in general the same as the CMC coefficient, and may even depend on the sample size. Simply plugging in the CMC $\hat{b}$ can actually degrade QMC's performance in some problems. Our

implementation did exactly estimate $\hat{b}$, the same way as we estimated $\hat{b}$ for CMC, so the CV is not tailored to the RQMC sampling scheme. Together with the already low variance of the RQMC estimator, this may explain why the VRFs is modest and sometimes below 1, despite the correlation between the control and target are close to 1.

Figure 6 further illustrates the trade-off when we fix the $\gamma = 1.2$, which is our main setup, and vary the number of paths. For small N the control variate can indeed reduce the RQMC standard error by a noticeable factor but the time ratio is also large because estimating $\hat{b}$ becomes relatively expensive. As a result the overall efficiency measure VRF/time ratio, rarely exceeds 1, only at $N = 16,384$. and $N = 131,072$ meaning that for most N 's, adding the CV will cost more in computational power than the reduction of variance.

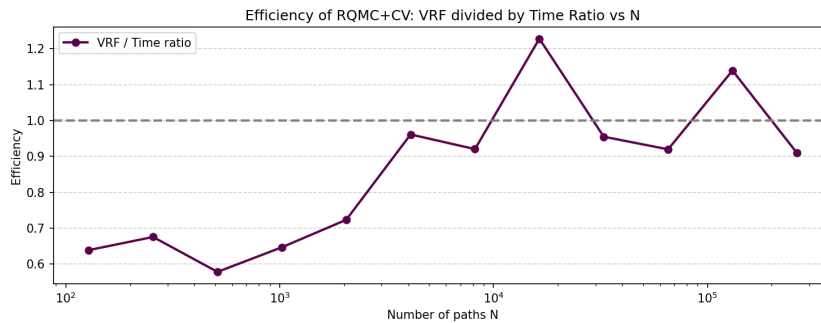


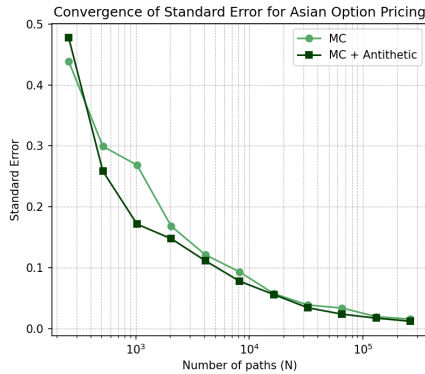
Figure 6: Control Variate RQMC: Variance Reduction Factor and Time ratio plotted for varying sample size N . Estimated with $S_0 = 100$, $K = 100$, $T = 1$, $\sigma = 0.3$, $\gamma = 1.2$ and 32 scrambles.

7.4.2 Effect of Antithetic Variates

We assess antithetic variates (AV) by pairing each simulated path with its mirror and averaging the two payoffs, keeping all other settings fixed. Results are reported for both Crude Monte Carlo and Randomized Quasi-Monte Carlo. When generating paths, we use only half of the nominal sample size N , since applying the antithetic technique effectively doubles the number of paths by adding their negatives.

Table 7b shows that AV reduces sampling noise for CMC across most sample sizes. The variance reduction factor (VRF) is typically above 1 and reaches its maximum at $N = 1,024$. The improvements are smaller for other sample sizes. A slight negative gain is observed only at $N = 256$, which is not unusual in Monte Carlo experiments. This happens because the payoff function does not always change smoothly in response to the random inputs that drive the simulation (the so-called “driving normals”). When this relationship is weak, the antithetic paths are not strongly negatively correlated, and the variance reduction becomes less effective. Importantly, the mean prices with and without

AV are statistically indistinguishable, confirming that AV preserves unbiasedness. The reported time ratios are close to 1 (slightly below in our runs), indicating a similar computational cost per effective sample. The reason for the computation time of AV is lower could be due to it takes longer to draw a path than taking the negative of an existing path. Overall, AV provides a simple and efficient variance reduction method for CMC. The size of the improvement mainly depends on how closely the payoff behaves in an opposite way when we negate the random shocks under the CEV dynamics.



(a)

N	Price _{cmc}	SE _{cmc}	Price _{Anti}	SE _{Anti}	VRF	Time ratio
256	17.5120	0.4388	17.7637	0.4776	0.8441	1.03
512	17.7987	0.2992	17.9615	0.2584	1.3407	1.01
1,024	17.7780	0.2684	17.9923	0.1714	2.4521	0.98
2,048	17.7410	0.1681	17.9698	0.1476	1.2971	0.97
4,096	17.7154	0.1212	17.8626	0.1113	1.1858	0.92
8,192	17.8635	0.0930	17.7973	0.0778	1.4289	0.92
16,384	17.8694	0.0574	17.8492	0.0558	1.0582	0.93
32,768	17.9363	0.0386	17.8370	0.0341	1.2813	0.94
65,536	17.9134	0.0337	17.8728	0.0236	2.0391	0.98
131,072	17.8790	0.0196	17.8700	0.0170	1.3293	0.96
262,144	17.8480	0.0154	17.8570	0.0119	1.6747	0.94

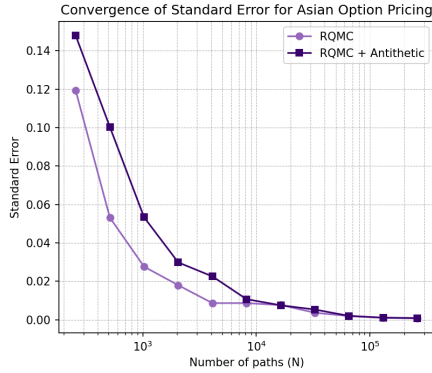
(b)

Figure 7: Antithetic CMC: Plot and Statistics

Estimated with $S_0 = 100$, $K = 100$, $T = 1$, $\sigma = 0.3$, $\gamma = 1.2$ and 32 scrambles.

In contrast, when AV is applied to RQMC (Table 8b), the improvements are often negative, except for when $N = 16,384$ where it only just passes 1 (1.0268) in VRF. For medium-sized N , the VRF values are in most cases below 1. This could be because of how the Brownian Bridge method is implemented. The Brownian Bridge already reduces most of the randomness by concentrating the largest part of the variance into the early dimensions of the Sobol' sequence. As a result, the paths generated are already highly correlated and well-structured, leaving little remaining "independent noise" for AV to cancel out. Only at the largest sample sizes do we see small improvements, and even then, the standard errors are already very small. As with CMC, the estimated option prices with and without AV are almost identical, showing that the method does not introduce bias. Considering that AV adds some extra implementation effort but gives only a very small or even negative effect, it does not provide a consistent advantage when RQMC with Brownian Bridge is already used.

Another reason for this difference in effectiveness between CMC and RQMC can be understood intuitively. Antithetic variates work best when the option payoff changes in a predictable and opposite way when the random inputs are negated. This creates a strong negative correlation between each path and its mirror, which reduces variability when the two are averaged. However, in our model, the payoff does not always move smoothly



(a)

N	Price _{CMC}	SE _{CMC}	Price _{Anti}	SE _{Anti}	VRF	Time ratio
256	17.8806	0.1192	17.8602	0.1478	0.6504	1.00
512	17.8535	0.0531	17.8816	0.1003	0.2803	1.03
1,024	17.8619	0.0276	17.8828	0.0533	0.2681	1.04
2,048	17.8293	0.0180	17.8728	0.0298	0.3648	1.03
4,096	17.8332	0.0087	17.8652	0.0226	0.1482	1.06
8,192	17.8378	0.0087	17.8514	0.0107	0.6611	1.05
16,384	17.8491	0.0076	17.8423	0.0075	1.0268	1.07
32,768	17.8489	0.0036	17.8487	0.0053	0.4614	1.05
65,536	17.8486	0.0019	17.8479	0.0020	0.9025	1.07
131,072	17.8461	0.0010	17.8487	0.0011	0.8264	1.05
262,144	17.8458	0.0007	17.8471	0.0009	0.6049	1.03

(b)

Figure 8: Antithetic RQMC: Plot and Statistics

Estimated with $S_0 = 100$, $K = 100$, $T = 1$, $\sigma = 0.3$, $\gamma = 1.2$ and 32 scrambles.

or in a perfectly opposite direction when we negate the random shocks. The Brownian Bridge technique further changes how the random numbers affect the path by reordering the importance of the random inputs, making the first few Sobol' numbers control most of the variation. Because of this, the antithetic paths are no longer as negatively correlated as before, and the benefit of AV becomes very limited.

A related explanation comes from Art B Owen (2008). The paper shows that variance improvements in RQMC can require local antithetic constructions tailored to the underlying digital net and to sufficiently smooth integrands. By contrast, the standard global mirroring used here may not align with the Sobol' structure, especially once the Brownian Bridge has concentrated most variation in the earliest dimensions. This may explain why AV provides little or even negative improvement in our RQMC results.

In summary, AV is a useful and easy variance reduction tool for CMC, but it adds no value when combined with RQMC and Brownian Bridge in this setting.

7.5 Combined Comparison of Accuracy and Efficiency

In this section we compare all Monte Carlo estimators and variance-reduction combinations simultaneously, in terms of both accuracy (standard error) and computational cost (runtime). The goal is to assess which methods provide the best trade-off between precision and efficiency when pricing the arithmetic Asian option.

In Figure 9, we have plotted the standard errors given the number of paths. As expected, CMC exhibits the highest standard errors across all sample sizes. Incorporating antithetic or control variates reduces the standard error, consistent with the results from Section 7.4. Moreover, combining control variates and antithetic variates within CMC provides

an additional improvement: the combined estimator consistently attains lower standard errors than using the control variate alone. This is consistent with the fact that the two variance-reduction techniques operate through different mechanisms—correlation reduction and regression adjustment—and therefore their effects are cumulative rather than redundant.

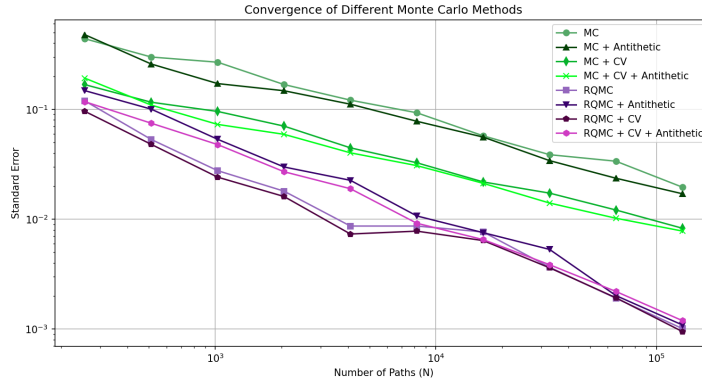


Figure 9: Plot of standard errors of all estimators against the number of paths.

Estimated with $S_0 = 100$, $K = 100$, $T = 1$, $\sigma = 0.3$, $\gamma = 1.2$ and 32 scrambles.

Next, we consider the RQMC-based methods. As previously shown, control variates reduce the variance of RQMC, while antithetic variates increase it, due to the diminished negative correlation introduced by the Brownian Bridge ordering. When combining the two, we therefore expect an intermediate effect. Indeed, the combined RQMC + CV + Antithetic estimator achieves a lower standard error than RQMC + Antithetic, but still performs worse than both RQMC and RQMC + CV sample sizes lower than 10^4 . As N gets larger all RQMC variants converge to similar (low) standard errors, although RQMC + CV + Antithetic remains the least efficient of the RQMC-based estimators.

While Figure 9 compares accuracy alone, efficiency also depends on computational time. Figure 10 therefore plots standard error against the realized runtime for each estimator. For the CMC methods, adding control variates is consistently preferable: both CMC + CV and CMC + CV + Antithetic lie below their counterparts with the same runtime. Antithetic variates also provide a cheap reduction in variance for CMC, as they add virtually no computational overhead.

For the RQMC methods, using the control variate substantially increases runtime, especially at larger sample sizes. Although RQMC + CV ultimately reaches the smallest standard errors, it is not the most efficient choice when only moderate precision is required. Only when we have time for a 5-6 second simulation, then adding the control variate make sense. As we allow for more runtime, all of the RQMC estimators eventually

converge to same, except for the RQMC+CV+Antithetic, which performs slightly worse.

So, depending on how much time you have to price the Asian option, different estimators is actually preferred. If you have a little less than a second to price the option, the CMC with control and antithetic variates is preferred. However if you have more time, and want a higher precision the RQMC is preferred. Adding the control variate to the RQMC estimator only makes sense if you have around 5-10 seconds to run the simulation. We also find that using antithetic variates for the RQMC is not worth the trouble. Numerical values supporting these conclusions are listed in Appendix B.1

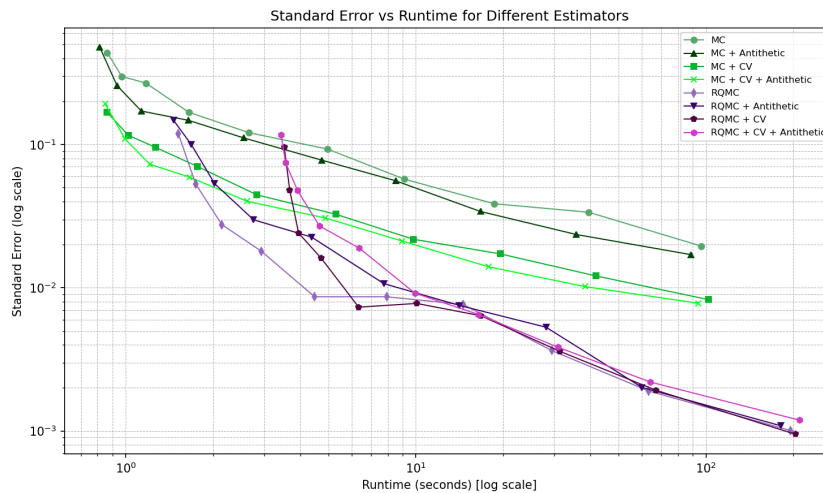


Figure 10: Every estimator plotted by their standard errors and their runtime.

Estimated with $S_0 = 100$, $K = 100$, $T = 1$, $\sigma = 0.3$, $\gamma = 1.2$ and 32 scrambles.

8 Sensitivity Analysis of Greek Estimators

In this section, we estimate the Delta and Vega of the arithmetic Asian option under the CEV model using Monte Carlo and randomized Quasi-Monte Carlo (RQMC) simulation. Since no analytical Greek formulas exist under CEV dynamics, and the pathwise derivation method does not apply, sensitivities must be approximated numerically. We employ the central finite-difference (central FD) estimator, combined with common random numbers (CRN) to reduce noise. We first examine the choice of perturbation size ϵ , and then analyse the convergence and variance behaviour of the resulting Greek estimators under CMC, RQMC, and variance-reduction techniques.

8.1 Step-size sensitivity

The accuracy of the finite-difference estimators depends on the bump size ϵ . Small bumps amplify Monte Carlo noise, while large bumps introduce finite-difference bias. To study this trade-off, we evaluate several relative bump sizes for Delta and Vega using common random numbers. Both Delta and Vega is estimated using the central difference method, as we assure that σ stays positive by using relative bumps. We estimate both Greeks using the previous estimation methods, except for the methods RQMC+Antithetic and RQMC+CV+Antithetic, as the introduction of Antithetic variates only introduced noise to the estimator.

Figure 11 shows the resulting Delta estimates and their standard errors for the different relative bumps. Across all methods, the Delta values remain highly stable for intermediate bump sizes (around 10^{-2}), but becomes unreliable for very large jumps and small jumps. For smaller bump sizes (small ϵ) the central finite difference estimator suffers from numerical cancellation. Although the central scheme removes first-order bias, it does not mitigate the numerical cancellation that arises when subtracting two Monte Carlo estimates that are nearly identical.

It is particularly noticeable for control-variate versions of the estimators, there seem to be some bias at bump sizes below 0.005. This behaviour arises because the regression coefficients used in the CV adjustment are estimated separately at $+\epsilon$ and $-\epsilon$. At very small bump sizes these estimates differ due to regression noise, and since the geometric-Asian control variate carries almost no finite-difference signal, the resulting asymmetry is amplified when divided by 2ϵ . Thus the apparent upward drift reflects noise amplification rather than genuine finite-difference bias. A formal derivation of this cancellation and its implications is given later. In contrast, RQMC exhibits smoother behaviour for small bumps because the quasi-random points create stronger correlation between the bumped and unbumped prices.

Vega remains stable across bump sizes from 10^{-4} to 10^{-2} , as shown in Figures 12a-b. A noticeable feature of the plot is that the CMC-based estimators with antithetic variates lie consistently above the RQMC estimators across all bump sizes. This difference is not caused by the bump size nor by the use of antithetic variates. Rather, it reflects a convergence effect: in the bump-size experiment we use $N = 16,384$ paths, and at this sample size the CMC Vega estimator has not yet stabilised. As seen later, CMC+antithetic Vega typically oscillates and tends to overestimate the true sensitivity at most N .

For bump sizes approaching unity, all methods exhibit a visible decline in Vega estimates. This reflects finite-difference bias: large perturbations move the volatility parameter far enough that the central-difference approximation no longer captures the local slope of the

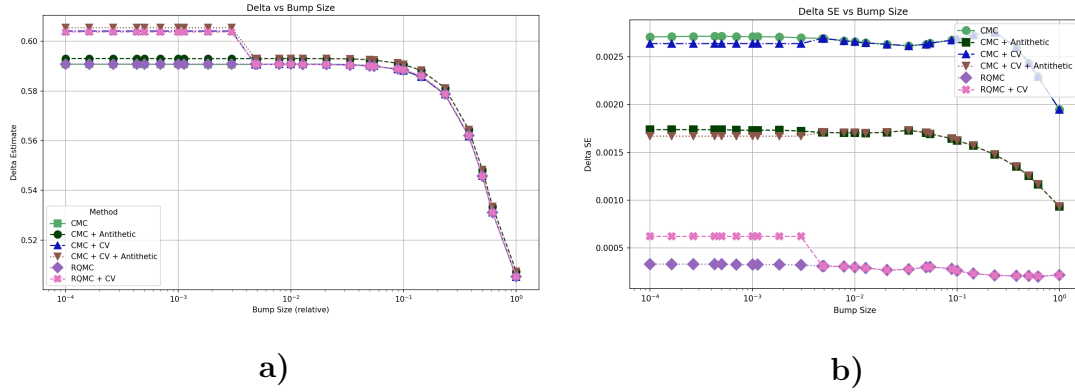


Figure 11: Delta Estimates (a) and Delta Standard Errors (b) for different relative perturbances
 Estimated with $N = 16,384$, $S_0 = 100$, $K = 100$, $T = 1$, $\sigma = 0.3$, $\gamma = 1.2$ and 32 scrambles

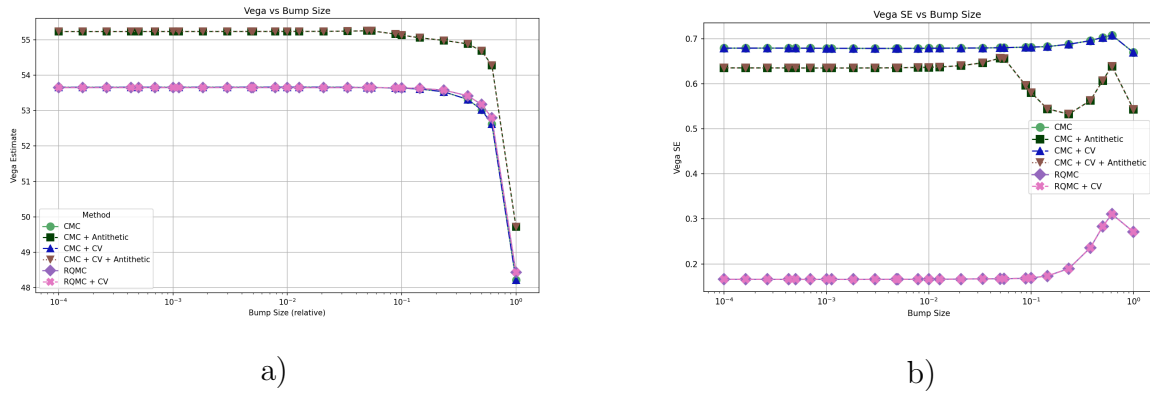


Figure 12: Vega Estimates (a) and Vega Standard Errors (b) for different relative perturbances
 Estimated with $N = 16,384$, $S_0 = 100$, $K = 100$, $T = 1$, $\sigma = 0.3$, $\gamma = 1.2$ and 32 scrambles

price with respect to σ . In this region, the correlation between the $+\epsilon$ and $-\epsilon$ evaluations also weakens, which causes the RQMC variance to increase.

Based on these experiments, we select a bump size of 10^{-2} for the remaining analysis, as this provides a good balance between variance and bias. Smaller bumps suffer from numerical cancellation and noise amplification, while larger bumps introduce finite-difference bias. In this intermediate range, both Delta and Vega remain stable across all estimators.

8.2 Variance and Convergence of Greek Estimators

With our chosen bump-size, we can now assess how the derivatives converge. In Figure 13 and 14, we have plotted the Delta and Vega estimates respectively, using the different estimators. The first thing to notice is that CMC oscillates the most for lower N 's. Greeks

are differences of two highly correlated Monte Carlo estimates, and small errors in each price estimate can amplify the variability of the Greek. As a result, CMC Greeks converge slowly and non-monotonically. RQMC converges faster and more smoothly. Even for small N the RQMC estimator produces tighter confidence intervals. This reflects the reduction in effective dimension achieved by combining Sobol points with the Brownian bridge construction. Overall, RQMC seem to clearly dominate CMC in terms of convergence speed and estimator stability for both Delta and Vega.

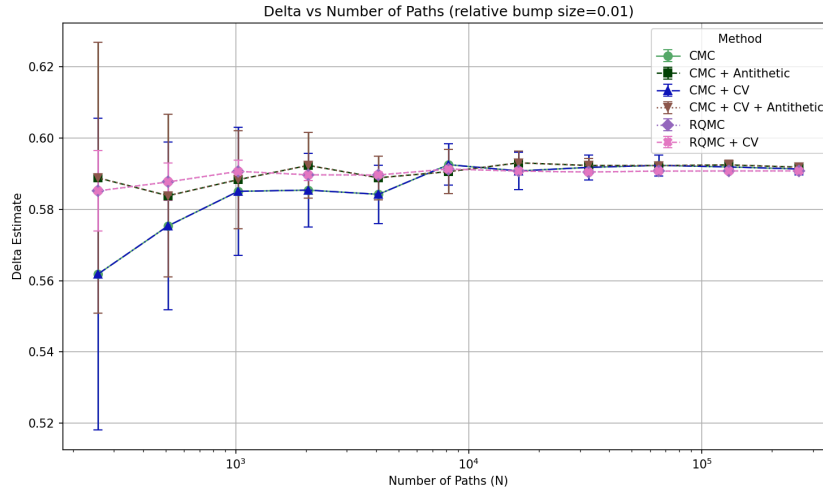


Figure 13: Delta estimates, with corresponding 95% confidence intervals

Estimated with $S_0 = 100$, $K = 100$, $T = 1$, $\sigma = 0.3$, $\gamma = 1.2$ and 32 scrambles.

A conspicuous result is that the control variate, in contrast to the pricing, does not improve the estimation of the Greeks. This is however consistent with the theory, as we can see by the FD estimator for the control variate:

$$\hat{\Delta}_{CV} = \frac{\hat{P}_{CV}(\theta + \epsilon) - \hat{P}_{CV}(\theta - \epsilon)}{2\epsilon}$$

If we then insert the control variate function

$$\begin{aligned} &= \frac{1}{2\epsilon} \left[\hat{P}(\theta + \epsilon) + b(X(\theta + \epsilon) - E[X(\theta + \epsilon)]) - \hat{P}(\theta - \epsilon) - b(X(\theta - \epsilon) - E[X(\theta - \epsilon)]) \right] \\ &= \frac{\hat{P}(\theta + \epsilon) - \hat{P}(\theta - \epsilon)}{2\epsilon} + \frac{b(X(\theta + \epsilon) - X(\theta - \epsilon) - E[X(\theta + \epsilon)] + E[X(\theta - \epsilon)])}{2\epsilon} \\ &= \Delta_{plain} + \frac{b(X(\theta + \epsilon) - X(\theta - \epsilon) - E[X(\theta + \epsilon)] + E[X(\theta - \epsilon)])}{2\epsilon} \end{aligned}$$

where Δ_{plain} is the central-difference estimator without control variate. Since we use common random numbers and a small bump ϵ , the Monte Carlo estimates $(\theta + \epsilon)$ and

$(\theta - \epsilon)$ are almost symmetrical, the adjustment term practically cancels out making the CV contribution negligible. The CV method is inherently ill-suited for finite-difference Greeks unless the control variate reflects the derivative of the payoff with respect to the parameter, which the geometric Asian payoff does not in our experiment.

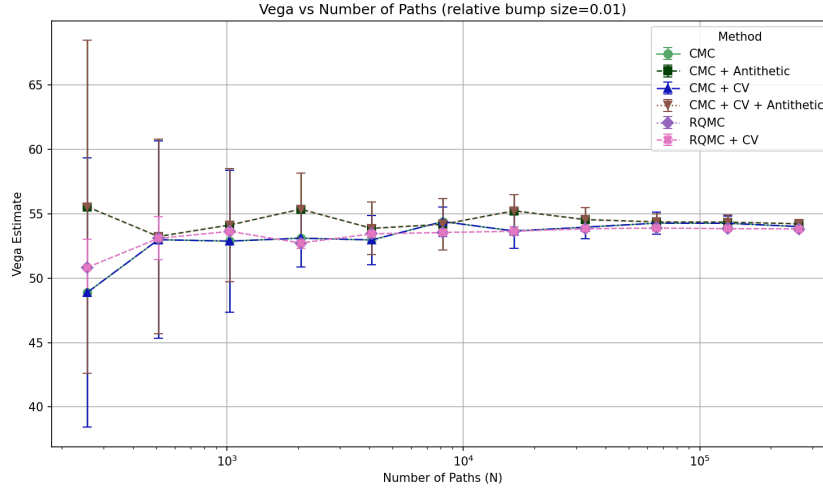


Figure 14: Vega estimates with corresponding 95% confidence intervals

Estimated with $S_0 = 100$, $K = 100$, $T = 1$, $\sigma = 0.3$, $\gamma = 1.2$ and 32 scrambles.

Antithetic variates, provides an improvement for Delta estimation, while the vega estimation seems to get a more modest reduction. This is expected as the delta estimator is moderately monotonic in the underlying Brownian increments, so negative correlation helps. Vega is however not monotonic in the randomness, especially with CEV dynamics, so antithetic do not systematically reduce variance. This is more prominent in the Appendix B.2, which shows a table of the delta and vega estimates by different number of paths, where the VRF for the Vega estimate actually dips below 1 multiple times.

These features are summarized numerically in Table 15, which reports Delta and standard errors for the largest sample size, $N = 262,144$. RQMC consistently gives a higher VRF than the antithetic variates but the effect on delta and Vega is not relatively consistent. Sometimes the VRF of using RQMC on Delta is higher while some other times VRF on Vega is higher. For $N = 262,144$, RQMC yields higher VRF for the Delta than Vega. The runtime does increase when using RQMC, but the reduction in variance outgrows the cost of the computation time.

These results illustrate some important insights for computing the sensitivities of the Asian option under the CEV model. The first thing is that Finite differences amplify Monte Carlo noise, and therefore Greeks benefit more from low discrepancy methods. The RQMC is superior for reducing variance for the Greeks but using Control variates is

ineffective when the control variate only mimics the payoff of the geometrics Asian option.

Method	Delta				Vega			
	Estimate	SE	VRF	Time (s)	Estimate	SE	VRF	Time (s)
CMC	0.591338	0.000871	1.00x	157	54.007450	0.199277	1.00x	161
CMC + Antithetic	0.591853	0.000515	2.86x	156	54.226210	0.169575	1.38x	160
CMC + CV	0.591338	0.000871	1.00x	165	54.007450	0.199277	1.00x	163
CMC + CV + Antithetic	0.591853	0.000515	2.86x	164	54.226210	0.169575	1.38x	160
RQMC	0.590803	0.000096	82.21x	211	53.832715	0.026358	57.16x	210
RQMC + CV	0.590803	0.000096	82.21x	217	53.832715	0.026358	57.16x	221

Figure 15: Summary of Delta and Vega estimates

Estimated with $S_0 = 100$, $K = 100$, $T = 1$, $\sigma = 0.3$, $\gamma = 1.2$, $N = 262,144$ and 32 scrambles.

9 Conclusion

In this paper, we examined the pricing of arithmetic Asian call options under the Constant Elasticity of variance model using Monte Carlo and randomized Quasi-Monte Carlo simulations. Our study sought to evaluate both pricing performance and sensitivity estimation using different techniques to reduce variance. However, all numerical results and performance comparisons in this study are conditional on the specific parameter configuration tested and should be interpreted as indicative rather than universally valid.

We firstly established that the Milstein discretization offers higher accuracy than Euler-Maruyama for simulating the CEV dynamics, justifying its use for the subsequent analysis. We then investigated the efficiency of the Crude Monte Carlo compared to RQMC with a Brownian bridge construction, which reduces the effective dimension of the problem. RQMC was found to clearly outperform CMC. Across all tested sample sizes, RQMC achieved variance reductions of up to two orders of magnitude relative to CMC, enabling more precise estimates for the same computational cost.

For the effectiveness of the variance reduction technique, we found that using the geometric Asian option under Black-Scholes, was effective with CMC, Especially when the CEV elasticity parameter γ is close to 1, where the dynamics resemble geometric Brownian motion. For our chosen $\gamma = 1.2$, the control variate delivered a meaningful reduction in variance at only a small increase in computational cost. The control variate was also able to decrease the variance for RQMC, but at a lower rate. Antithetic variates, however, provided only modest improvements for CMC and were consistently counterproductive when combined with RQMC. This behaviour aligns with theoretical expectations, as the Brownian Bridge reshapes the dependence structure in a way that weakens the negative correlation that antithetic sampling relies upon.

In the sensitivity analysis, we found that Greeks estimated via finite differences are significantly more noise-sensitive than price estimates. RQMC was especially advantageous here, providing large variance reductions for both Delta and Vega, while control variates performed poorly due to regression noise when bump sizes were small. Our results underscore that accurate Greek estimation under CEV dynamics requires both careful selection of the finite-difference bump size and the use of low-discrepancy sampling.

Overall, our findings show that randomized Quasi-Monte Carlo with Brownian Bridge construction is the most efficient and robust method for pricing arithmetic Asian options under CEV dynamics, and it is especially effective for computing sensitivities. Control variates enhance CMC performance but add little to RQMC in this context, whereas antithetic variates are generally unhelpful for RQMC. These insights highlight that the optimal simulation method depends heavily on both the computational budget and the precision requirements. When high accuracy is required, RQMC is clearly superior; when time is extremely limited, CMC with simple variance reductions may still be competitive. Future research could extend these techniques to multi-factor CEV models or explore alternative variance-reduction strategies better aligned with low-discrepancy sampling.

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A Appendix

A.1 Algorithm for generating a single path of the CEV price process

Algorithm 1 Generating a single path of the CEV price process

```

1: Input:  $S_0, T, m, \sigma, \gamma, r, \Delta W = (\Delta W_1, \dots, \Delta W_{2^m})$ 
2:  $M \leftarrow 2^m$ 
3:  $\tilde{X}_0 \leftarrow \log(S_0)$ 
4:  $\Delta t \leftarrow \frac{T}{M}$ 
5: for  $j = 1, \dots, M$  do
6:   if updating with Euler then
7:      $\tilde{X}_j \leftarrow \tilde{X}_{j-1} + (r - \frac{1}{2}\sigma^2 e^{2(\gamma-1)\tilde{X}_{j-1}})\Delta t + \sigma e^{(\gamma-1)\tilde{X}_{j-1}}\Delta W_j$ 
8:   else if updating with Milstein then
9:      $\tilde{X}_j \leftarrow \tilde{X}_{j-1} + (r - \frac{1}{2}\sigma^2 e^{2(\gamma-1)\tilde{X}_{j-1}})\Delta t + \sigma e^{(\gamma-1)\tilde{X}_{j-1}}\Delta W_j + \frac{1}{2}\sigma^2(\gamma - 1)e^{2(\gamma-1)\tilde{X}_{j-1}}((\Delta W_j)^2 - \Delta t)$ 
10:   end if
11:    $S_j \leftarrow \exp(\tilde{X}_j)$ 
12: end for
13: return  $(S_0, \dots, S_M)$ 

```

Note. ΔW is given as input, e.g. the standard Brownian increments or by Brownian bridge construction (Algorithm 2).

A.2 Algorithm for Brownian Bridge Construction

Algorithm 2 Brownian Bridge Construction

```

1: Input:  $m, (t_1, \dots, t_{2^m})$ 
2:  $M \leftarrow 2^m$ 
3: Generate Scrambled Sobol' point  $(U_1, \dots, U_M) \in [0, 1)^M$ 
4: for  $i = 1, \dots, M$  do
5:    $Y_i \leftarrow \Phi^{-1}(U_i)$ 
6: end for
7:  $h \leftarrow M, \quad j_{max} \leftarrow 1$ 
8:  $w_h \leftarrow \sqrt{t_h} Y_h$ 
9:  $t_0 \leftarrow 0, \quad w_0 \leftarrow 0$ 
10: for  $k = 1, \dots, m$  do
11:    $i_{min} \leftarrow \frac{h}{2}, \quad i \leftarrow i_{min}$ 
12:    $l \leftarrow 0, \quad r \leftarrow h$ 
13:   for  $j = 1, \dots, j_{max}$  do
14:      $a \leftarrow \frac{(t_r - t_i)w_l + (t_i - t_l)w_r}{t_r - t_l}$ 
15:      $b \leftarrow \sqrt{\frac{(t_i - t_l)(t_r - t_i)}{t_r - t_l}}$ 
16:      $w_i \leftarrow a + bY_i$ 
17:      $i \leftarrow i + h, \quad l \leftarrow l + h, \quad r \leftarrow r + h$ 
18:   end for
19:    $j_{max} \leftarrow 2 \cdot j_{max}$ 
20:    $h \leftarrow i_{min}$ 
21: end for
22: for  $i = 1, \dots, M$  do
23:    $\Delta W_i \leftarrow w_i - w_{i-1}$ 
24: end for
25: return  $(\Delta W_1, \dots, \Delta W_M)$ 

```

A.3 Algorithm for Pricing Arithmetic Asian Option

Algorithm 3 Pricing Arithmetic Asian Option

```

1: Input:  $S_0, T, m, N, K, R, r, \sigma, \gamma$ 
2:  $M \leftarrow 2^m$ 
3:  $\Delta t \leftarrow \frac{T}{M}$ 
4: for  $s = 1, \dots, R$  do
5:   for  $n = 1, \dots, N$  do
6:     Generate  $\Delta W = (\Delta W_1, \dots, \Delta W_M)$  using Algorithm 2 or  $\Delta W_i = \sqrt{\Delta t} Z_i$  with
        $Z_i \sim i.i.d.N(0, 1)$ 
7:     Simulate a path  $(S_n^s) = (S_{n,1}^s, \dots, S_{n,M}^s)$  using Algorithm 1
8:      $\bar{S}_n^s \leftarrow \frac{1}{M} \sum_{i=1}^M S_{n,i}^s$ 
9:      $\hat{P}_n^s \leftarrow \exp(-r \cdot T) \cdot \max(\bar{S}_n^s - K, 0)$ 
10:   end for
11:    $\hat{P}^s \leftarrow \frac{1}{N} \sum_{n=1}^N \hat{P}_n^s$ 
12: end for
13:  $\hat{P} \leftarrow \frac{1}{R} \sum_{s=1}^R \hat{P}^s$ 
14: return  $\hat{P}$ 

```

B Appendix

B.1 Estimation Results

N	Method	Price	SE	Time(s)	VRF (compared to CMC)
256	CMC	17.511980	0.438828	0.622603	1.000000
	CMC + Antithetic	17.763656	0.477567	0.640976	0.844345
	CMC + CV	17.691233	0.167707	0.675541	6.846767
	CMC + CV + Antithetic	17.774938	0.192067	0.666646	5.220184
	RQMC	17.880562	0.119211	1.136219	13.550534
	RQMC + Antithetic	17.860158	0.147849	1.115116	8.809581
	RQMC + CV	17.890006	0.095835	2.799482	20.967107
	RQMC + CV + Antithetic	17.899433	0.117205	2.662147	14.018287
512	CMC	17.798676	0.299177	0.770031	1.000000
	CMC + Antithetic	17.961489	0.258399	0.742003	1.340521
	CMC + CV	17.928196	0.115993	0.799989	6.652586
	CMC + CV + Antithetic	17.872750	0.109221	0.778701	7.503177
	RQMC	17.853475	0.053135	1.303502	31.703079
	RQMC + Antithetic	17.881627	0.100302	1.283498	8.896767
	RQMC + CV	17.862447	0.047949	2.881098	38.930454
	RQMC + CV + Antithetic	17.900194	0.074672	2.766546	16.052528
1024	CMC	17.778046	0.268424	0.933039	1.000000
	CMC + Antithetic	17.992278	0.171428	0.899021	2.451764
	CMC + CV	17.869582	0.095585	0.995084	7.886055
	CMC + CV + Antithetic	17.901312	0.072853	0.954436	13.575178
	RQMC	17.861946	0.027637	1.588973	94.335683
	RQMC + Antithetic	17.882837	0.053332	1.560096	25.331870
	RQMC + CV	17.867257	0.024120	3.047235	123.852852
	RQMC + CV + Antithetic	17.895976	0.047738	3.057621	31.616859
2048	CMC	17.741042	0.168071	1.288869	1.000000
	CMC + Antithetic	17.969801	0.147582	1.235358	1.296939
	CMC + CV	17.858110	0.070295	1.372923	5.716538
	CMC + CV + Antithetic	17.876096	0.059183	1.294698	8.064736
	RQMC	17.829255	0.017978	2.155008	87.397418
	RQMC + Antithetic	17.872763	0.029842	2.104874	31.719913
	RQMC + CV	17.833028	0.016073	3.662173	109.344841
	RQMC + CV + Antithetic	17.878127	0.027022	3.681578	38.684561
4096	CMC	17.715375	0.121160	2.107015	1.000000
	CMC + Antithetic	17.862644	0.111271	1.937784	1.185650
	CMC + CV	17.825440	0.044582	2.219710	7.385741
	CMC + CV + Antithetic	17.854305	0.040155	2.166246	9.103932
	RQMC	17.833175	0.008651	3.499671	196.135571
	RQMC + Antithetic	17.865231	0.022565	3.463299	28.828976
	RQMC + CV	17.840439	0.007321	4.946089	273.924539
	RQMC + CV + Antithetic	17.866451	0.018921	5.047011	41.005402
8192	CMC	17.653491	0.093012	3.951618	1.000000
	CMC + Antithetic	17.797306	0.077787	3.805777	1.429758
	CMC + CV	17.834767	0.032600	4.153522	8.140428
	CMC + CV + Antithetic	17.843698	0.030712	3.871374	9.172129
	RQMC	17.837770	0.008653	6.175182	115.553246
	RQMC + Antithetic	17.851447	0.010695	6.175503	75.633674
	RQMC + CV	17.841971	0.007775	7.844705	143.098274
	RQMC + CV + Antithetic	17.856979	0.009177	7.877396	102.735988
16384	CMC	17.869376	0.057364	7.211357	1.000000
	CMC + Antithetic	17.849195	0.055846	6.768937	1.055082
	CMC + CV	17.827609	0.021814	7.674802	6.915126
	CMC + CV + Antithetic	17.841937	0.021196	6.984576	7.324403
	RQMC	17.849087	0.007641	11.215492	56.362080
	RQMC + Antithetic	17.842290	0.007531	11.185858	58.014763
	RQMC + CV	17.851687	0.006381	13.039149	80.818314
	RQMC + CV + Antithetic	17.846124	0.006509	13.071731	77.672873
32768	CMC	17.936329	0.038625	14.731666	1.000000
	CMC + Antithetic	17.837024	0.034095	13.503156	1.283382
	CMC + CV	17.850041	0.017252	15.673815	5.012844
	CMC + CV + Antithetic	17.830733	0.014016	14.465317	7.593878
	RQMC	17.848893	0.003648	22.452156	112.080512
	RQMC + Antithetic	17.848684	0.005301	22.730197	53.087538
	RQMC + CV	17.849346	0.003605	25.246569	114.790388
	RQMC + CV + Antithetic	17.851231	0.003846	24.974378	100.843805
65536	CMC	17.913426	0.033685	35.374087	1.000000
	CMC + Antithetic	17.872816	0.023568	30.350247	2.042833
	CMC + CV	17.854869	0.012108	34.644310	7.739172
	CMC + CV + Antithetic	17.850917	0.010189	31.984071	10.929541
	RQMC	17.848634	0.001904	51.358059	313.155854
	RQMC + Antithetic	17.847888	0.002014	51.721597	279.848682
	RQMC + CV	17.848215	0.001923	54.888525	306.944549
	RQMC + CV + Antithetic	17.848289	0.002195	54.987792	235.523202
131072	CMC	17.879001	0.019559	89.044551	1.000000
	CMC + Antithetic	17.869990	0.017032	76.849933	1.318809
	CMC + CV	17.846638	0.008266	86.003691	5.599082
	CMC + CV + Antithetic	17.850791	0.007790	81.530116	6.303664
	RQMC	17.846096	0.001009	170.980197	376.059815
	RQMC + Antithetic	17.848714	0.001088	160.404682	323.170913
	RQMC + CV	17.846309	0.000948	168.364945	425.658728
	RQMC + CV + Antithetic	17.848288	0.001192	169.857924	269.453287
262144	CMC	17.848005	0.015407	188.762524	1.000000
	CMC + Antithetic	17.857036	0.011908	184.170156	1.673928
	CMC + CV	17.847095	0.005532	209.902157	7.758011
	CMC + CV + Antithetic	17.850307	0.005216	201.399024	8.726631
	RQMC	17.845849	0.000692	393.876315	496.278559
	RQMC + Antithetic	17.847102	0.000875	387.969867	310.315414
	RQMC + CV	17.845756	0.000721	402.755411	456.076403
	RQMC + CV + Antithetic	17.847267	0.000995	419.030968	239.818669

Table 5: Price estimates for all methods across different number of paths

B.2 Greek estimates

N	Method	Delta Estimate	Delta SE	Delta VRF	Delta Time (s)	Vega Estimate	Vega SE	Vega VRF	Vega Time (s)
256	CMC	0.561823	0.022348	1.000000	0.571000	48.886972	5.340903	1.000000	0.790000
	CMC + Antithetic	0.588861	0.019394	1.330000	0.573000	55.547642	6.594971	0.660000	0.589000
	RQMC	0.585188	0.005781	14.940000	0.699000	50.833362	1.127061	22.460000	0.700000
512	CMC	0.575363	0.012020	1.000000	0.705000	52.991199	3.905362	1.000000	0.932000
	CMC + Antithetic	0.583842	0.011644	1.070000	0.668000	53.240776	3.845458	1.030000	0.674000
	RQMC	0.587710	0.002684	20.060000	0.805000	53.090122	0.851147	21.050000	0.794000
1024	CMC	0.585069	0.009202	1.000000	0.816000	52.877993	2.817557	1.000000	1.063000
	CMC + Antithetic	0.588346	0.007011	1.720000	0.800000	54.124506	2.248010	1.570000	0.813000
	RQMC	0.590684	0.001635	31.690000	0.986000	53.636860	0.379928	55.000000	0.955000
2048	CMC	0.585411	0.005275	1.000000	1.102000	53.108636	1.133231	1.000000	1.452000
	CMC + Antithetic	0.592355	0.004723	1.250000	1.081000	55.359071	1.424768	0.630000	1.145000
	RQMC	0.589699	0.000802	43.290000	1.310000	52.736352	0.219259	26.710000	1.291000
4096	CMC	0.584224	0.004202	1.000000	1.767000	52.968742	0.975056	1.000000	2.343000
	CMC + Antithetic	0.588871	0.003123	1.810000	1.705000	53.872524	1.041544	0.880000	1.844000
	RQMC	0.589658	0.000456	85.000000	2.067000	53.446065	0.158658	37.770000	2.063000
8192	CMC	0.592559	0.002960	1.000000	3.219000	54.395607	0.580366	1.000000	4.144000
	CMC + Antithetic	0.590589	0.003176	0.870000	3.102000	54.186142	1.014089	0.330000	3.187000
	RQMC	0.591333	0.000298	98.390000	3.815000	53.547671	0.128048	20.540000	3.793000
16384	CMC	0.590787	0.002658	1.000000	5.860000	53.663136	0.678560	1.000000	7.598000
	CMC + Antithetic	0.593071	0.001704	2.430000	5.705000	55.234583	0.636312	1.140000	6.118000
	RQMC	0.590742	0.000299	78.780000	6.910000	53.648579	0.165834	16.740000	6.894000
32768	CMC	0.591791	0.001776	1.000000	11.955000	53.961219	0.449937	1.000000	13.343000
	CMC + Antithetic	0.592324	0.001034	2.950000	11.460000	54.551872	0.477631	0.890000	11.995000
	RQMC	0.590491	0.000312	32.310000	14.026000	53.838839	0.064437	48.760000	13.871000
65536	CMC	0.592364	0.001508	1.000000	26.619000	54.275182	0.442983	1.000000	27.529000
	CMC + Antithetic	0.592298	0.000464	10.550000	25.476000	54.369631	0.328592	1.820000	27.130000
	RQMC	0.590755	0.000229	43.290000	31.302000	53.889007	0.097024	20.850000	30.852000
131072	CMC	0.591908	0.000919	1.000000	66.223000	54.262820	0.288463	1.000000	68.827000
	CMC + Antithetic	0.592488	0.000652	1.990000	64.326000	54.353287	0.286178	1.020000	70.430000
	RQMC	0.590792	0.000140	42.930000	86.792000	53.846886	0.039339	53.770000	87.418000
262144	CMC	0.591338	0.000871	1.000000	157.370000	54.007450	0.199277	1.000000	160.511000
	CMC + Antithetic	0.591853	0.000515	2.860000	156.217000	54.226210	0.169575	1.380000	159.761000
	RQMC	0.590803	0.000096	82.210000	211.246000	53.832715	0.026358	57.160000	210.284000

Table 6: Greek estimations. Control variate methods is excluded as there is very little to no effect on the estimation